Reservoir / ESN realisations of signature volatility models

Nonlinear readouts, structured vector fields, and the controlled-equation umbrella — with the Volterra-kernel / truncated-signature equivalence made precise

July 2, 2026

Contents

*Working note. The aim is to expand three structural observations — (a) nonlinear (quadratic) readouts, (b) structured rather than random vector fields, (c) the random controlled differential equation (RCDE) umbrella — and, as the technical spine, to state exactly in what sense the **Volterra kernels** that appear in the reservoir-computing universality proofs are **equivalent to the truncated signature**. Every object is defined; theorems are stated with references; a closing section proposes results that look provable, separating the known from the open.*

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1. Purpose and the one-paragraph thesis

The signature volatility models of Cuchiero–Gazzani–Svaluto-Ferro and Abi Jaber–Gérard, and the randomized-signature / reservoir-computing programme of Cuchiero–Gonon–Grigoryeva–Ortega–Teichmann (CGGOT), are **the same model class viewed from two traditions**. In both, the instantaneous volatility is a *readout* of a high-dimensional process generated by an *exogenous* driving path; calibration trains only the readout while the generating dynamics stay fixed. The mathematical glue is classical: a causal, time-invariant, fading-memory input/output map admits a **Volterra-series** approximation (Boyd–Chua), the Volterra series of a *state-affine* system is precisely a **linear functional of the path signature** (Chen–Fliess), and a **random projection** of that signature is the CGGOT randomized signature, which is an **echo state network (ESN)**. Consequently:

> The exact signature volatility model (Abi Jaber–Gérard) is the **state-affine, linear-readout corner** of the ESN family; CGGOT is its **dimension-reduced random** version. Viewing the construction as an ESN exposes three handles the signature presentation leaves on the table — **nonlinear readouts (a)**, **structured vector fields (b)**, and an **input-driven/exogenous dial (c)** — each backed by existing reservoir-computing approximation theory, and each relevant to reproducing path-dependent stylised facts (notably the strong Zumbach effect) while preserving the simulate-once / re-fit-cheaply property.

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2. Standing notation and definitions

Throughout, $d \in \mathbb{N}$ is the number of driving channels, $N \in \mathbb{N}$ the reservoir dimension, and $T > 0$ a finite horizon.

Definition 2.1 (Driving path; time augmentation). Let $W = (W^1, \dots, W^{d-1})$ be a standard $(d-1)$-dimensional Brownian motion. The **time-extended** (or time-augmented) path is

$$\widehat{W}_t \;=\; (t,\, W_t^1, \dots, W_t^{d-1}) \;\in\; \mathbb{R}^d, \qquad \widehat{W}_t^0 := t. \tag{2.1}$$

Time augmentation is not cosmetic: the 0-th (time) channel is what produces the *mean-reversion / exponential-decay* structure in the kernels below. We write X for a generic continuous semimartingale driver in $\mathbb{R}^d$ (e.g. $X = \widehat{W}$), and use Stratonovich integration ($\circ\, dX$) so that the chain rule, hence Chen's identity, holds pathwise.

The asset price is driven by a **scalar** Brownian motion Z, $dS_t/S_t = \sigma_t\, dZ_t$, correlated to the reservoir noise through a leverage vector $\varrho \in \mathbb{R}^{d-1}$, $\|\varrho\| \le 1$:

$$dZ_t = \varrho^\top dW_t + \sqrt{1 - \|\varrho\|^2}\, dB_t^\perp, \qquad B^\perp \perp W, \;\; B^\perp \text{ scalar}. \tag{2.2}$$

Thus Z is the leverage projection of W plus one idiosyncratic degree of freedom $B^{\perp}$; it is not an independent d-th noise. When the reservoir is to be given access to the price-driving direction (§6–§7), one augments the control to (t, W, Z) — equivalently $(t, W, B^{\perp})$ — at the cost of a single extra scalar channel.

Definition 2.2 (Signature and truncated signature). The **signature** of X over $[0, t]$ is the tensor-algebra element

$$\mathbb{S}(X)_{0,t} \;=\; \Big(1,\ \mathbb{X}^{(1)}_{0,t},\ \mathbb{X}^{(2)}_{0,t}, \dots\Big) \;\in\; T(\!(\mathbb{R}^d)\!) := \prod_{k\geq 0} (\mathbb{R}^d)^{\otimes k}, \tag{2.3}$$

with k-th level

$$\mathbb{X}^{(k)}_{0,t} = \int_{0<s_1<\cdots<s_k<t} \circ\, dX_{s_1} \otimes \cdots \otimes \circ\, dX_{s_k}, \qquad \mathbb{X}^{i_1\cdots i_k}_{0,t} = \int_{0<s_1<\cdots<s_k<t} \circ\, dX^{i_1}_{s_1} \cdots \circ\ dX^{i_k}_{s_k}. \tag{2.4}$$

For a word (multi-index) $I = (i_1, \dots, i_k) \in \{0, \dots, d-1\}^k$ we write $\langle \mathbb{S}(X)_{0,t}, I\rangle := \mathbb{X}^I_{0,t}$ for the corresponding coordinate. The **truncated signature of order** M is the projection

$$\mathbb{S}^{\leq M}(X)_{0,t} \in T^M := \bigoplus_{k=0}^{M} (\mathbb{R}^d)^{\otimes k}, \qquad \dim T^M = \sum_{k=0}^{M} d^k = \frac{d^{\,M+1}-1}{d-1}. \tag{2.5}$$

The signature solves the **universal (Chen) controlled differential equation (CDE)**

$$d\,\mathbb{S}(X)_{0,t} = \mathbb{S}(X)_{0,t} \otimes \circ\, dX_t, \qquad \mathbb{S}(X)_{0,0} = \mathbf{1}. \tag{2.6}$$

Definition 2.3 (Shuffle product). For words I, J, the coordinate functions multiply by the **shuffle product** $ш$:

$$\langle \mathbb{S}, I\rangle\, \langle \mathbb{S}, J\rangle = \langle \mathbb{S},\ I \,ш\, J\rangle, \tag{2.7}$$

where $I \,ш\, J$ is the formal sum of all interleavings of I and J preserving internal order. Hence **products of linear functionals of the signature are again linear functionals of the signature** (of higher degree). This algebraic fact is used heavily in (a).

Definition 2.4 (Fading memory property, FMP). Let K_M be a space of (left-infinite) input sequences/paths bounded by M, equipped with a weighted norm $\|u\|_w = \sup_{t\leq 0} w(-t)\,|u_t|$ for a strictly decreasing weight $w \downarrow 0$. A causal, time-invariant operator (filter) $\mathcal{F} : K_M \to \mathbb{R}$ has the **fading memory property** if it is continuous w.r.t. $\|\cdot\|_w$: inputs that agree on the recent past and differ only far in the past produce nearly equal outputs. FMP is the natural domain of both Volterra-series and reservoir approximation. It is stated *before* the reservoir/ESN (Definition 2.6) deliberately: the **echo state property** defined there is exactly FMP transported from the input/output map to the internal state recursion ("ESP = FMP for the state dynamics"), so FMP must be available first; and FMP is the common target class of the entire §3 approximation chain (FMP filter → Volterra series → signature → randomized signature).

Definition 2.5 (Volterra series). A (finite, order-M) **Volterra series** operator acting on an input $x : (-\infty, t] \to \mathbb{R}^{d-1}$ is

$$\mathcal{V}_M[x]_t = h_0 + \sum_{k=1}^{M} \sum_{i_1,\dots,i_k} \int_0^{\infty} \cdots \int_0^{\infty} h_k^{i_1\cdots i_k}(\tau_1, \dots, \tau_k) \prod_{r=1}^{k} dx^{i_r}_{t-\tau_r}, \tag{2.8}$$

with **Volterra kernels** h_k. The diagonal of h_2 encodes ARCH-type clustering; the **off-diagonal, time-asymmetric part of** h_2 encodes lead–lag / trend effects (this is the object the strong Zumbach effect lives in, §6–§8).

Definition 2.6 (Reservoir system / ESN; state-affine system). A **reservoir system** is a state-space filter

$$R_n = F(R_{n-1}, x_n), \qquad y_n = h(R_n), \qquad R_n \in \mathbb{R}^N, \tag{2.9}$$

with F the (fixed, often random) reservoir map and h the trained readout. The continuous-time analogue is a controlled differential equation

$$dR_t = V_0(R_t)\,dt + \sum_{i=1}^{d-1} V_i(R_t) \circ dX_t^i, \qquad y_t = h(R_t). \tag{2.10}$$

The system is **state-affine (SAS) / bilinear** if the vector fields are linear in the state,

$$V_0(r) = A_0\,r, \qquad V_i(r) = A_i\,r \quad (A_0, A_i \in \mathbb{R}^{N\times N}), \tag{2.11}$$

and the readout is linear, $h(r) = w^\top r$. An **echo state network** is the special discrete-time case $F(r, x) = \sigma(\mathbf{A}r + \mathbf{C}x + \zeta)$ with σ a componentwise activation and $(\mathbf{A}, \mathbf{C}, \zeta)$ fixed/random. The system has the **echo state property (ESP)** if for each input there is a unique bounded state sequence independent of initialisation (the dynamical-systems form of FMP; a contractivity/Lyapunov condition on $\mathbf{A}$).

Definition 2.7 (Randomized signature, CGGOT). Fix N and draw random $A_i \in \mathbb{R}^{N\times N}$, $b_i \in \mathbb{R}^N$ $(i = 0, \dots, d-1)$ and an activation σ. The **randomized signature** is the $\mathbb{R}^N$-valued process

$$dR_t = \sum_{i=0}^{d-1} \sigma\big(A_i R_t + b_i\big) \circ dX_t^i, \qquad R_0 = r_0. \tag{2.12}$$

With $\sigma = \mathrm{id}$ and $b_i = 0$ this is a state-affine system (2.11); with the canonical shift operators A_i it *is* the genuine signature (2.6) in coordinates. CGGOT show R_t is a random, dimension-N projection of $\mathbb{S}(X)_{0,t}$ whose linear span reproduces the signature's approximation power.

Definition 2.8 (Signature volatility model; Abi Jaber–Gérard, Cuchiero et al.). A **signature volatility model** sets the spot volatility to a (possibly infinite) **linear functional of the signature** of $\widehat{W}$,

$$\sigma_t = \langle \ell,\ \mathbb{S}(\widehat{W})_{0,t}\rangle = \sum_I \ell_I\,\mathbb{X}^I_{0,t}, \qquad \frac{dS_t}{S_t} = \sigma_t\,dZ_t, \quad d\langle Z, W^i\rangle_t = \varrho_i\,dt, \tag{2.13}$$

with leverage vector ϱ, $\|\varrho\| \le 1$. Truncating ℓ at $|I| \le M$ gives the truncated-signature model. By Definition 2.6 this is **exactly the state-affine, linear-readout reservoir** (2.11) driven by $\widehat{W}$. (Convention: Abi Jaber–Gérard place the linear functional on the *volatility* σ_t; from §5 onward we place the readout on the *variance* $V_t = \sigma_t^2$ instead — the natural target for a quadratic, strictly-positive readout — so "readout" means the map $R_t \mapsto V_t$ there. Nothing downstream depends on which of σ or V is designated, only on it being a signature/reservoir functional.)

Remark 2.9 (how the SAS and the ESN relate). Definitions 2.6–2.7 name *one* reservoir template at two levels of nonlinearity, and (2.12) is the dial between them. Setting $\sigma = \mathrm{id}$, $b_i = 0$ in (2.12) gives $dR_t = \sum_i A_i R_t \circ dX_t^i$ — the **state-affine / bilinear** system (2.11), linear in the state with the input entering multiplicatively — which, with the canonical shift operators, *is* the exact signature (2.6). Switching on a nonlinear σ gives the **ESN**. Hence the SAS is precisely the *linear-activation corner* of the ESN; conversely, a first-order Taylor expansion of the activation, $\sigma(z) = \sigma(\zeta) + \sigma'(\zeta)(z - \zeta) + \cdots$, reduces an ESN's update to an affine (state-affine) one — the SAS is the ESN's linearisation. Both are reservoir systems (2.9) and both are FMP-universal

(Theorem 3.7): they are the *same object at two levels of nonlinearity*, which is exactly why the signature/Volterra equivalence is **exact** on the SAS corner and holds only **to the order of the expansion** once σ is nonlinear (Remark 3.4), and why the exact signature model (Definition 2.8) is "the state-affine corner of the ESN family" (§1).

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3. The spine: Volterra kernels are the resummed truncated signature

This section makes precise the sentence "the Volterra kernel used in the randomized-signature proof is equivalent to the truncated signature." The cleanest equivalence is **exact for state-affine reservoirs** and is the backbone of the signature-volatility corner; the randomized (nonlinear) case is a controlled generalisation, addressed in Remark 3.4.

What is and is not new here. None of the individual results is new: Lemma 3.1 is Chen–Fliess (1957–1981), and CGGOT already prove that the randomized signature is a universal *random projection* of the signature. What this section isolates is the **explicit, exact kernel-level** form of that correspondence in the state-affine corner — the matrix-exponential resummation (3.4) and the crisp "Volterra order = signature level" of (3.5) — and uses it as the bridge that identifies the signature-volatility models (Abi Jaber–Gérard, CGS) with the reservoir / randomized-signature programme (CGGOT) as *one* class. CGGOT establish the *approximate, random, nonlinear* side (universality at rate $O(N^{-1/2})$); the exactly-solvable *linear* corner and its closed kernel are what make the fractional/rough kernel structure (Corollary 3.3) and the exact-vs-approximate split (Remark 3.4) legible. So the contribution is synthesis and precision, not a new universality theorem (this is why T1 is flagged "known" in §8).

3.1 Chen–Fliess expansion of a state-affine reservoir

Lemma 3.1 (Chen–Fliess / Picard expansion). For the state-affine system $dR_t = \sum_{i=0}^{d-1} A_i R_t \circ dX_t^i$, $R_0 = r_0$, the solution is the everywhere-convergent series

$$R_t = \sum_{k\geq 0} \sum_{I=(i_1,\ldots,i_k)} \left(A_{i_k} A_{i_{k-1}} \cdots A_{i_1}\right) r_0 \ \langle \mathbb{S}(X)_{0,t}, I\rangle. \tag{3.1}$$

Hence any linear readout is a **linear functional of the signature**,

$$y_t = w^\top R_t = \sum_{k\geq 0} \sum_I \underbrace{\left(w^\top A_{i_k} \cdots A_{i_1} r_0\right)}_{=:\,\ell_I} \ \mathbb{X}^I_{0,t} = \langle \ell, \mathbb{S}(X)_{0,t}\rangle, \tag{3.2}$$

and the **order-M truncation of** (3.2) **is the truncated signature** $\mathbb{S}^{\leq M}$ paired with the coefficients ℓ_I. The initial state r_0 has not disappeared: it is the rightmost factor of every coefficient $\ell_I = w^\top A_{i_k} \cdots A_{i_1} r_0$ (with $\ell_\emptyset = w^\top r_0$ the constant term $k = 0$), i.e. it is absorbed into the linear functional ℓ. The triple $(w, \{A_i\}, r_0)$ is thus an over-parametrisation of a single ℓ; the signature-volatility model (Definition 2.8) simply parametrises ℓ directly.

Proof. Iterate $R_t = r_0 + \sum_i A_i \int_0^t R_s \circ dX_s^i$. The depth-$k$ term is $\sum_{i_1,\ldots,i_k} A_{i_k} \cdots A_{i_1} r_0 \int_{0<s_1<\cdots<s_k<t} \circ dX_{s_1}^{i_1} \cdots \circ dX_{s_k}^{i_k}$; the inner integral is $\mathbb{X}^I_{0,t}$ by Definition 2.2. Convergence on $[0,T]$ follows from the factorial decay $\|\mathbb{X}^{(k)}_{0,t}\| \leq \|X\|^k_{1\text{-var};[0,t]}/k!$ (bounded-variation case; the rough-path estimate in the Brownian case is analogous). ■

3.2 Resummation of the time channel produces exponential Volterra kernels

Split the drift channel ($i = 0$, $dX^0 = dt$, write $A_0 =: A$) from the stochastic channels. Resumming **all insertions of the time channel** between consecutive stochastic increments turns the signature series (3.2) into a Volterra series (2.8) with explicit **matrix-exponential kernels**.

Theorem 3.2 (Volterra–signature equivalence for state-affine reservoirs). Let $dR_t = AR_t\,dt + \sum_{i=1}^{d-1} B_i R_t \circ dX_t^i$ with $A := A_0$, $B_i := A_i$ (Stratonovich, per Definition 2.1). Then the linear readout $y_t = w^\top R_t$ admits the **finite-or-infinite Volterra representation**

$$y_t = \sum_{k\geq 0} \sum_{i_1,\ldots,i_k} \int_{0<s_1<\cdots<s_k<t} K^{(k)}_{i_1\cdots i_k}(t; s_1, \ldots, s_k)\ dX^{i_1}_{s_1} \cdots dX^{i_k}_{s_k}, \tag{3.3}$$

with **separable exponential kernels**

$$K^{(k)}_{i_1\cdots i_k}(t; s_1, \ldots, s_k) = w^\top e^{A(t-s_k)} B_{i_k}\, e^{A(s_k-s_{k-1})} B_{i_{k-1}} \cdots e^{A(s_2-s_1)} B_{i_1}\, e^{As_1}\, r_0. \tag{3.4}$$

Equivalently, $K^{(k)}$ is exactly the **resummation over the time-channel of the order-k (in the stochastic channels) signature coefficients** of Lemma 3.1: expanding each $e^{A\tau} = \sum_{m\geq 0} A^m \tau^m/m!$ and using $\int_{\text{simplex}} \prod \tau_r^{m_r}$ = (products of time-channel iterated integrals) reproduces (3.2) term by term, where each factor $A^m\tau^m/m!$ corresponds to m insertions of the time word 0. In particular,

$$\boxed{\text{truncating the Volterra order at } M \iff \text{truncating the signature at level } M \text{ in the stochastic channels.}} \tag{3.5}$$

Proof sketch. In (3.1) group words I by their subsequence of **nonzero** letters $(i_1, \ldots, i_k)$ and sum over all numbers and positions of interleaved zeros. A maximal run of p zeros between the $(r-1)$-th and r-th nonzero increments contributes $\int_{s_{r-1}<u_1<\cdots<u_p<s_r} du_1 \cdots du_p \cdot A^p = \frac{(s_r-s_{r-1})^p}{p!} A^p$ acting on the left of $B_{i_{r-1}}$; summing $p \geq 0$ gives $e^{A(s_r-s_{r-1})}$. The boundary runs give $e^{A(t-s_k)}$ on the left and e^{As_1} on the right, yielding (3.4). ■

Corollary 3.3 (the kernels live in the "rough"/multi-scale class). If $A = -\mathrm{diag}(\kappa_1, \ldots, \kappa_N)$ then the level-1 kernel is $K^{(1)}_i(t; s) = w^\top e^{A(t-s)} B_i\, e^{As} r_0 = \sum_n w_n\, e^{-\kappa_n(t-s)}\, (B_i\, e^{As} r_0)_n$; **as a function of the lag $t - s$ it is a finite sum of exponentials** with rates κ_n (the factor $e^{As} r_0$ is the fading initial-condition transient — absent if the reservoir is started in the infinite past, where the kernel depends on the lag alone). This is the standard multi-factor Markovian approximation of a (rough) Volterra kernel K_H; choosing κ_n on a log-grid with loadings $\propto \kappa_n^{1/2-H}$ targets $K_H(\tau) = \tau^{H-1/2}/\Gamma(H+1/2)$ via its Laplace representation $K_H(\tau) = \int_0^\infty e^{-\kappa\tau} \mu_H(d\kappa)$. (This is the precise sense in which "fractional loading" is a kernel-quadrature weight, and why a finite reservoir is only *pre-asymptotically* rough.)

3.3 The universality chain, and where the randomized signature enters

Putting the pieces together yields the chain that underlies the CGGOT and ESN universality theorems:

$$\underbrace{\text{FMP filter } \mathcal{F}}_{\text{target}} \xrightarrow[\text{Boyd–Chua}]{\text{Thm 3.5}} \underbrace{\text{finite Volterra } \mathcal{V}_M}_{(2.8)} \xrightarrow[\text{Thm 3.2}]{\text{exact, SAS}} \underbrace{\langle \ell, \mathbb{S}^{\leq M} \rangle}_{\text{truncated signature}} \xrightarrow[\text{random projection}]{\text{Thm 3.6, CGGOT}} \underbrace{\text{randomized signature}}_{(2.12)} \tag{3.6}$$

Direction of approximation. The arrows run from the *rich target on the left* to the *finite, tractable realisation on the right*. The object we care about — the true volatility functional, a fading-memory filter of the driving path (and any rough / path-dependent behaviour we want it to exhibit) — is the target $\mathcal{F}$; the finite-N ESN / randomized signature is the **working model that approximates it**, with error $O(N^{-1/2})$ from the random projection (and a truncation error from cutting the signature at level M). So the honest statement is not "we posit a Volterra financial model and then approximate it by an ESN," but "we *adopt* the ESN as the tractable, Markovian, simulate-once model, and the universality chain *certifies* that it approximates the whole FMP/Volterra/signature class (rough-vol, PDV, signature-vol) as $N \to \infty$." In the state-affine corner (middle arrow) there is no approximation at all — the signature-volatility model *is* an ESN exactly (Theorem 3.2); approximation enters only through truncation (M) and the random projection (N).

Theorem 3.5 (Boyd–Chua, 1985). Any causal, time-invariant operator with the fading memory property on a uniformly bounded, equicontinuous set of inputs can be approximated **uniformly** by a finite Volterra series (2.8); equivalently, by a finite-dimensional state-space system with a polynomial (analytic) readout.

Theorem 3.6 (Randomized-signature universality and rate; CGGOT 2021, Gonon–Grigoryeva–Ortega 2020/2023). Let $\mathcal{F}$ be FMP. For the randomized signature (2.12) of dimension N there exists a linear readout $w \in \mathbb{R}^N$ such that the resulting filter approximates $\mathcal{F}$ in L^2, with error $O(N^{-1/2})$ under the random-feature/Monte-Carlo concentration governing the projection (Johnson–Lindenstrauss-type control of the signature coordinates). The reservoir (A_i, b_i, σ) is **fixed** and **input-independent**; only w is trained.

Theorem 3.7 (ESN / SAS universality; Grigoryeva–Ortega 2018, Gonon–Ortega 2020/2021). ESNs with linear readouts, and non-homogeneous state-affine systems with linear readouts, are **universal** in the FMP class, including for **stochastic inputs** (Gonon–Ortega 2020). The reservoir supplies the Volterra/signature feature expansion (3.3); the linear readout selects the kernel.

Theorem 3.8 (Volterra reservoir kernel; Gonon–Grigoryeva–Ortega). There is a universal kernel — the **Volterra reservoir kernel** — whose sections approximate any FMP filter; it is built from the reservoir feature map of a state-space realisation of the Volterra-series expansion. This is the kernel-method incarnation of (3.3) and the operational form of "Volterra series $\equiv$ reservoir features."

Remark 3.4 (exact vs approximate equivalence — the precision point). The equivalence (3.5) is **exact** when the vector fields are *linear in the state* (state-affine / bilinear, Theorem 3.2), which is precisely the Abi Jaber–Gérard signature-volatility setting (Definition 2.8): there, $\sigma_t = \langle \ell, \mathbb{S}(\widehat{W})_{0,t} \rangle$ *is* a state-affine reservoir readout, and its Volterra kernels are (3.4). When CGGOT switch on a **nonlinearity** $\sigma \neq \mathrm{id}$ in (2.12), the reservoir is no longer state-affine: its Chen–Fliess/Volterra expansion still exists (Boyd–Chua for analytic σ; Fliess series), but the kernels are no longer the bare (3.4) — they are the **Taylor-resummed** kernels of the nonlinear flow, and the identity "Volterra order = signature level" holds **to the order of the expansion**, not verbatim. The randomized signature is therefore best read as: *a finite-dimensional random reservoir whose linear span L^2-approximates the same FMP/Volterra/signature class*, with the **state-affine case the exactly-solvable corner**. This distinction matters when one wants closed-form (Fourier/Riccati) pricing — that tractability is a property of the *linear/affine* corner, not of the general ESN.

3.4 Relation to Volterra-process (rough-volatility) models

Orientation. Rough-volatility models are built on a **Volterra process**; §3.1–§3.3 are about a **Volterra series**. These are different objects. Once they are separated, the reservoir picture turns out to reproduce the standard rough-volatility construction *exactly* in one regime and to make transparent what it may and may not claim in the others. A reader who wants only the upshot can jump to the dictionary in Step 3.

Step 1 — two different objects, both named "Volterra." In a rough-volatility model the variance is a **Volterra process**: it solves a stochastic Volterra equation (SVE) in which the fractional kernel is convolved into the *dynamics*,

$$V_t = V_0 + \int_0^t K(t-s)\, b(V_s)\, ds + \int_0^t K(t-s)\, \sigma(V_s)\, dW_s, \qquad K(t) = \frac{t^{H-1/2}}{\Gamma(H+\frac{1}{2})}, \quad H \in (0, \tfrac{1}{2}], \tag{3.7}$$

the affine-Volterra / rough-Heston class (Abi Jaber–Larsson–Pulido; El Euch–Rosenbaum). The object of §3 is instead a **Volterra series** (Definition 2.5), in which the kernel sits in the *output map*: $V_t = h_0 + \sum_k \int \cdots \int h_k \prod_r dW_{t-\tau_r}$. The two are related but distinct. Solving the SVE by Picard iteration and expanding in W *does* produce a Volterra series, but its kernels $\{h_k\}$ are built from *iterated convolutions* of the dynamical kernel K (with $h_1 \propto K$, and each further h_k carrying one more convolution of K weighted by derivatives of b, σ); so "the kernel K of the SVE" and "the kernels h_k of the series" are genuinely different objects, and §3 speaks about the latter. Keeping them apart is what makes the statements below exact rather than merely suggestive.

Step 2 — a rough kernel is a continuous average of exponentials, and truncating that average is the reservoir. The single bridge between the two worlds is a classical fact: for $H \in (0, \frac{1}{2})$ the kernel $K_H(t) = t^{H-1/2}/\Gamma(H + \frac{1}{2})$ is *completely monotone*, so by **Bernstein's theorem** it is the Laplace transform of a positive measure,

$$K_H(t) = \int_0^\infty e^{-xt}\, \mu_H(dx), \qquad \mu_H(dx) = \frac{x^{-H-1/2}}{\Gamma(H+\frac{1}{2})\,\Gamma(\frac{1}{2}-H)}\, dx. \tag{3.8}$$

Approximating μ_H by a finite sum of point masses $\sum_{n=1}^N c_n\, \delta_{\kappa_n}$ (a quadrature, $c_n \geq 0$) yields a proxy kernel $K_N(t) = \sum_n c_n e^{-\kappa_n t}$, and each exponential $e^{-\kappa_n t}$ is the kernel of an Ornstein–Uhlenbeck factor $Y_t^n = \int_0^t e^{-\kappa_n(t-s)}\, dW_s$. If every factor is driven by the **same** Brownian motion W (this hypothesis is essential — see caveat (ii)), put $Z_t^N := \sum_n c_n Y_t^n = \int_0^t K_N(t-s)\, dW_s$. Then, by Itô's isometry, for the Gaussian Volterra process $Z_t := \int_0^t K_H(t-s)\, dW_s$,

$$\sup_{t \leq T} \mathbb{E}\big|Z_t^N - Z_t\big|^2 = \sup_{t \leq T} \int_0^t \big|K_N - K_H\big|^2(t-s)\, ds \;\leq\; \|K_N - K_H\|^2_{L^2(0,T)}, \tag{3.9}$$

so **whenever the quadrature converges in** L^2 — $\|K_N - K_H\|_{L^2(0,T)} \to 0$ — one has $Z^N \to Z$ in $L^2(\Omega)$, uniformly on $[0, T]$. Now $R^N := (Y^1, \ldots, Y^N)^\top$ is *exactly* the diagonal state-affine reservoir of Theorem 3.2 — drift $A = -\mathrm{diag}(\kappa_1, \ldots, \kappa_N)$, common driver — and $Z^N = c^\top R^N$ is a *linear readout*; moreover Corollary 3.3's fractional loading $c_n \propto \kappa_n^{1/2-H}$ (log-grid) is precisely the quadrature weight for μ_H. In one sentence: **the reservoir is the multifactor Markovian approximation of a rough kernel, realised at the level of the readout feature** — Carmona–Coutin–Montseny for fractional Brownian motion, Abi Jaber–El Euch for rough Heston — and it inherits their convergence theory, including the node-placement rates of Bayer–Breneis and Alfonsi–Kebaier (geometric in N for optimised / Gaussian-quadrature nodes, algebraic for a naïve log-grid).

Step 3 — a dictionary: the degree of the readout selects the model. With the reservoir fixed, climbing in readout degree walks through the standard models:

1. **Linear readout $\Rightarrow$ Gaussian Volterra process.** The driver $Z_t = \int_0^t K_H(t-s)\, dW_s$ (Riemann–Liouville fractional Brownian motion — the forward-variance driver of rough Bergomi

and the base state of quadratic rough Heston) is a single level-1 kernel, hence exactly the $N \to \infty$ limit of the linear readout $c^\top R^N$ of Step 2. This is the Gaussian/affine corner, and the SVE-side face of Remark 3.4's "exact for state-affine."

2. **Quadratic readout ⇒ quadratic rough Heston.** Taking $V_t = a\,(Z_t - b)^2 + c$ is the model of Gatheral–Jusselin–Rosenbaum, whose defining feature is a price-feedback (Zumbach) effect and which resolves the joint SPX/VIX smile calibration. It is a *quadratic readout on a (rough) Volterra state* — precisely the object of §5 — and $(Z-b)^2$ is the sign-folding of §4. The present construction is its finite-dimensional, Markovian, trainable reservoir generalisation.
3. **Exponential readout ⇒ (rough) Bergomi.** Taking $V_t = \xi_0 \exp\big(b^\top R_t - \frac{1}{2} b^\top P b\big)$ makes the variance *lognormal*, driven by the Gaussian factor $b^\top R_t$: this is the N-factor Bergomi model, and with Corollary 3.3's fractional loading it is the Markovian approximation of *rough Bergomi* (Bayer–Friz–Gatheral). Where item 1 furnishes Bergomi's Gaussian *driver*, the exponential readout furnishes the lognormal variance itself. It is the group-like resummation of the quadratic readout (§5, Prop. 5.3) and reproduces the same area-Zumbach mechanism (Corollary 9.9), trading the affine-VIX^2 tractability of the quadratic case (§9.1) for Bergomi's lognormal-basket VIX and a more delicate martingale property (§5; T7).
4. **Affine-but-nonlinear, and Markovian polynomial ⇒ rough Heston and polynomial-OU.** Rough Heston itself is an *affine* Volterra process with a genuinely nonlinear SVE (the $\sqrt{V}$ feedback): it is *not* a linear readout of a Gaussian reservoir, but is tractable through its fractional Riccati–Volterra characteristic function (El Euch–Rosenbaum) and is approximated by the nonlinear multifactor system of Abi Jaber–El Euch (a common-driver OU reservoir with square-root diffusion feedback). Dropping roughness but keeping a polynomial readout on a low-dimensional OU state gives the Gaussian-polynomial family — the quintic-OU and Gaussian polynomial models of Abi Jaber–Illand–Li, which also jointly calibrate SPX and VIX. All three are the *same* reservoir template at different kernel resolutions and readout degrees.

Three caveats, stated precisely.

- **(i) A finite reservoir is only *pre-asymptotically* rough.** Write the reservoir in Itô form $dR_t = b(R_t)\,dt + \Sigma(R_t)\,dW_t$, so $\Sigma(R_t)$ is its diffusion matrix (the coefficient of dW_t; constant Σ in the linear-Gaussian case of §9.1, state-dependent in general — this is the same Σ whose reversibility condition appears in §9.2). For fixed N, Z^N — and any C^2 readout $V = g(R^N)$ — is then a semimartingale with absolutely continuous quadratic variation, so, by Itô's lemma ($\|\Sigma^\top \nabla g\|^2 = \nabla g^\top \Sigma\Sigma^\top \nabla g$ being the instantaneous variance rate of V),

$$\lim_{\Delta \downarrow 0} \frac{1}{\Delta}\,\mathbb{E}\big[(V_{t+\Delta} - V_t)^2 \mid \mathcal{F}_t\big] = \big\|\Sigma(R_t)^\top \nabla g(R_t)\big\|^2 < \infty, \tag{3.10}$$

  i.e. the small-lag structure function scales like Δ^1 and the *infinitesimal* Hurst exponent is $\frac{1}{2}$. Rough scaling $m_2(\Delta) \propto \Delta^{2H}$ holds only over the intermediate band $\Delta \in [\kappa_{\max}^{-1}, \kappa_{\min}^{-1}]$ and becomes exact only in the limit $N \to \infty$, $\kappa_{\max} \to \infty$. (The true Volterra process with $H < \frac{1}{2}$ is neither Markovian nor a semimartingale — which is *why* a lift is needed at all.)
- **(ii) Pathwise roughness needs a *common* driver.** The mean-square convergence of Step 2 used the *same* W for every factor. If instead each factor carries its own *independent* Brownian, $\widetilde{Z}_t^N = \sum_n c_n \int_0^t e^{-\kappa_n (t-s)}\,dW_s^n$, then $\widetilde{Z}^N$ does **not** converge to Z; only its stationary autocovariance

$$\operatorname{Cov}\big(\widetilde{Z}_t^N, \widetilde{Z}_{t+\tau}^N\big) = \sum_n c_n^2\,\frac{e^{-\kappa_n|\tau|}}{2\kappa_n} \tag{3.11}$$

can be fitted to a power law — the Barndorff-Nielsen–Shephard superposition-of-OU long memory, a *second-order / in-distribution* effect, not a pathwise approximation. So the common-vs-independent-driver choice decides whether "rough" means *pathwise* or merely *covariance-level* — a distinction easily lost, and directly relevant to diagonal-reservoir architectures.
- **(iii) Well-posedness and martingality are separate questions.** Existence and uniqueness for the SVE (Abi Jaber–Larsson–Pulido) and the true-martingale property of the discounted price (Gassiat, for rough Bergomi; cf. T7) are not delivered by the approximation and must be established in their own right.

Positioning. Read this way, rough (Volterra), path-dependent (PDV), and Markovian-polynomial volatility models form *one* family — readouts of a common finite-dimensional reservoir, distinguished by the kernel quadrature (how rough) and the readout degree (how nonlinear). This is exactly the axis of Abi Jaber–Li ("Volatility models in practice: rough, path-dependent, or Markovian?"), and it is consistent with the empirical finding (Rømer) that a low-dimensional Markovian multi-OU model can match rough models on the joint calibration task. The reservoir view does not settle that debate; it makes the three paradigms commensurable inside a single object.

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4. Roadmap: how the three handles bear on the strong Zumbach effect

Before developing the handles in turn, here is, in one place, what each is *for*. The strong Zumbach effect — past trends forecasting future volatility more strongly than the reverse — is a **time-reversal asymmetry** whose arrow runs from returns to variance, and reproducing it needs two logically separate ingredients: a **magnitude (sign-folding) response** — volatility reacts to the *size* of past moves, not their sign, produced by an operation that is *even* in sign (the archetype is $x \mapsto x^2$, which "folds" $+x$ and $-x$ onto the same output); and a genuine **arrow of time** in the joint (return, variance) law. The three handles supply exactly these, plus control of *fidelity*:

- **(a) Nonlinear (quadratic) readouts (§5)** provide the sign-folding. A quadratic form in the reservoir turns a signed trend coordinate into a trend-*magnitude* term ($T \mapsto T^2$) — the ARCH / volatility-clustering mechanism. This is necessary but, on its own, time-*symmetric*: it is not yet the asymmetry (Corollary 5.2).
- **(b) Structured vector fields (§6)** provide the arrow of time, and do so interpretably. Placing *named* Lévy-area coordinates $\mathbb{A}^{Z,n}$ — the exactly antisymmetric, time-reversal-*odd* part of the signature (Lemma 6.2) — into the state loads precisely the coordinates that carry the asymmetry, instead of hoping a generic random reservoir smears it across all N.
- **(c) The exogenous-vs-price-driven dial (§7)** sets the *fidelity* — how faithfully the reproduced effect reflects the genuine mechanism (volatility as a functional of the realised price path) rather than a statistically matched surrogate. With an exogenous control that merely *includes* the price-driver Z, the effect is a reusable **proxy** (simulate once, re-fit cheaply); genuinely price-driving the reservoir gives the faithful strong effect, at the cost of that reusability.

The three are complementary, not competing: (a) folds sign into magnitude, (b) injects the time-irreversible content, (c) fixes how faithfully that content tracks the realised path. What binds them is one necessary condition — with *no* price→variance channel the asymmetry vanishes identically, for **any** readout (Proposition 9.4) — so a quadratic readout or an area coordinate yields a Zumbach signal only once such a channel (leverage, or Z in the control) is present. The linear-Gaussian benchmark of §9.1 makes all of this explicitly computable, and the central quantitative claim — that loading the area coordinates reproduces the strong-effect diagnostic at leading order — is stated as Conjecture 9.6.

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5. (a) Linear readout → nonlinear (quadratic and exponential) readouts

The signature models keep the readout linear by design, because affine structure is what gives the Fourier/Riccati transform of $(\log S, \int V)$ and hence semi-closed-form pricing (Abi Jaber–Gérard). ESN universality (Theorems 3.6–3.7) requires **no such restriction**, and modern reservoir computing routinely uses polynomial/quadratic readouts. Two precise statements justify moving to a quadratic readout for the *variance*.

Quadratic readout: the sign-folding second-order kernel

Proposition 5.1 (a quadratic readout on a level-M reservoir is a linear readout on a level-$2M$ signature). Let R_t be a state-affine reservoir whose readouts realise signature functionals up to level M (Lemma 3.1). For symmetric $Q \in \mathbb{R}^{N\times N}$, $w \in \mathbb{R}^N$, $c_0 \in \mathbb{R}$, the quadratic readout

$$V_t = c_0 + w^\top R_t + R_t^\top Q\, R_t \tag{5.1}$$

equals a **linear** functional of the signature **up to level** $2M$. Writing the state's signature series (Lemma 3.1) with its *vector* coefficients, $R_t = \sum_I M_I\, \mathbb{X}^I_{0,t}$ with $M_I := A_{i_k}\cdots A_{i_1} r_0 \in \mathbb{R}^N$ (each $\mathbb{X}^I_{0,t}$ a scalar), and substituting into the bilinear form,

$$R_t^\top Q R_t = \sum_{I,J} (\ell^Q_{I,J})\, \mathbb{X}^I_{0,t}\, \mathbb{X}^J_{0,t} = \sum_{I,J} \ell^Q_{I,J}\, \langle \mathbb{S},\ I \sqcup\!\sqcup J\rangle, \qquad \ell^Q_{I,J} := M_I^\top Q\, M_J, \tag{5.2}$$

i.e. the first equality collects the scalar coefficient $\ell^Q_{I,J} = M_I^\top Q M_J$ of each product $\mathbb{X}^I\mathbb{X}^J$, and the second is the shuffle identity (2.7). Hence (5.1) stays inside the signature-linear class **on the full signature**, but at **fixed reservoir dimension** N it is strictly more expressive than $w^\top R_t$: it accesses degree-$2M$ words a degree-M linear readout cannot.

Corollary 5.2 (quadratic readout = explicit second-order Volterra kernel = *sign-folding* engine). Substituting (3.3) into (5.1), the variance acquires a **genuine second-order Volterra kernel** in the stochastic channels:

$$V_t = c_0 + \underbrace{\textstyle\int K^{(1)}\, dX}_{\text{level-1}} + \underbrace{\textstyle\iint \widetilde{K}^{(2)}(t; s_1, s_2)\, dX_{s_1} dX_{s_2}}_{\text{from } R^\top QR \text{ and level-2 of } R}. \tag{5.3}$$

What the quadratic readout supplies is **sign-folding / magnitude sensitivity**, and it is important to see that this is the *symmetric* part of $\widetilde{K}^{(2)}$, not the time-asymmetric one. Concretely, a single causal trend coordinate $T_t = \int_0^t e^{-\kappa(t-s)} \circ dX_s$ (a level-1 readout) entering quadratically gives

$$T_t^2 = 2\int_{s_1<s_2} e^{-\kappa(t-s_1)}\, e^{-\kappa(t-s_2)} \circ dX_{s_1} \circ dX_{s_2}, \tag{5.4}$$

a kernel **symmetric** in (s_1, s_2): *a sustained move of either sign raises* V — the volatility-clustering / ARCH mechanism. By itself this is a magnitude effect and, as a kernel, carries **no** time-reversal asymmetry (contrast Definition 2.5 and Lemma 6.2, where the asymmetry lives in the *antisymmetric* part).

Where the Zumbach asymmetry then comes from — and why the readout alone is not enough. The quadratic readout is necessary (to fold sign into magnitude) but not sufficient for the asymmetry. Two routes make $Z(\tau) \neq 0$, and both require a genuine price→vol channel: 1. **Causal price-trend in the control.** If the trend T is built from the **price-driving direction** Z (i.e. Z is a reservoir input, §7), then even though the variance *process* $V = c_0 + T^2$ is itself time-reversible (an instantaneous function of a reversible OU), the *joint* (S, V) law acquires an arrow of time: the same Z drives the price and, through the past-weighted trend, the future variance, so the four-point functional $Z(\tau)$ becomes nonzero. This is the reusable **proxy** for the strong effect (§9.3), and it is exactly the Z-in-control case in which Prop. 9.4 does *not* force $Z \equiv 0$. 2. **Area coordinates.** The time-reversal asymmetry of the variance *process itself* is carried by the **antisymmetric / area** part of $\widetilde{K}^{(2)}$ (Lemma 6.2), which the quadratic readout reaches only when the reservoir carries area coordinates (non-commuting B_i, §6).

So the honest reading — consistent with §7 (row 3, "strong-*like*") and §9.3 — is: a quadratic readout on a *fixed exogenous* reservoir manufactures the sign-folding and, provided the price direction Z is in the control, a **reusable proxy** for the strong Zumbach effect; it does not require feeding back the realised S-path, but it does require the exogenous Z-channel, and with no price→vol channel at all Prop. 9.4 gives $Z(\tau) \equiv 0$ regardless of Q.

Theory that survives the move. The readout is still a finite-dimensional regression on the (now quadratic) feature map $r \mapsto (1, r, \mathrm{vech}(rr^\top))$; the Gonon–Grigoryeva–Ortega random-feature bounds (Theorem 3.6) apply verbatim to this enlarged but still finite feature set, giving the same $O(N^{-1/2})$ control. **What is lost** is the affine transform property: under a quadratic readout the pair $(\log S, \int_0^\cdot V)$ is no longer affine, so Fourier/Riccati pricing (Abi Jaber–Gérard) is replaced by Monte-Carlo on the stored reservoir paths. This is the explicit fidelity/tractability trade and should be stated as such in any write-up.

Exponential readout: the lognormal (Bergomi) corner

Proposition 5.3 (exponential readout: lognormal variance, and the full-signature character). The strictly positive exponential readout

$$V_t = \xi_0 \, \exp\!\big(b^\top R_t - \tfrac{1}{2}\, b^\top P b\big), \qquad \mathbb{E}[V_t] = \xi_0, \tag{5.5}$$

sharpens the quadratic case in three ways. **(1) Full-signature linearity.** The shuffle identity gives $\langle \ell, \mathbb{S}\rangle^k = \langle \ell^{\sqcup\!\sqcup k}, \mathbb{S}\rangle$, hence

$$e^{\langle \ell, \mathbb{S}\rangle} = \big\langle \exp_{\sqcup\!\sqcup}(\ell),\ \mathbb{S}\big\rangle, \qquad \exp_{\sqcup\!\sqcup}(\ell) := \sum_{k \geq 0} \tfrac{1}{k!}\, \ell^{\sqcup\!\sqcup k}: \tag{5.6}$$

the exponential readout is *again a linear functional of the signature*, now of the **full** signature with coefficient the shuffle-exponential — the group-like / character element of the shuffle Hopf algebra. The quadratic readout (5.1) is exactly its **second-order truncation** $\frac{1}{2}\, \ell^{\sqcup\!\sqcup 2}$; the exponential *resums* all shuffle powers. **(2) Unconditional positivity**, with no $Q \succeq 0$ constraint and no capping (contrast (5.1), which is positive only if $Q \succeq 0$ and c_0 dominates). **(3) Lognormal variance.** V_t is lognormal in the multi-factor OU $b^\top R_t$ — the N-factor Bergomi model, and with Corollary 3.3's fractional loading the Markovian approximation of rough Bergomi (§3.4, item 3). Its second-order structure is the closed-form lognormal one,

$$\operatorname{Cov}(V_t, V_{t+\tau}) = \xi_0^2\big(e^{c(\tau)} - 1\big), \qquad c(\tau) = b^\top P\, e^{A^\top \tau} b, \tag{5.7}$$

whose *log*-variance autocovariance $c(\tau)$ is exactly the multi-factor-OU sum of Prop. 9.1 (Bergomi's kernel).

Theory that survives the move. Even more directly than for the quadratic readout: since $\log V_t = b^\top R_t - \frac{1}{2} b^\top P b$ is *itself a linear readout* of the reservoir, the entire linear-readout apparatus transfers verbatim to the log-variance — the Gonon–Grigoryeva–Ortega random-feature bounds (Theorem 3.6) apply to $\log V$ with the same $O(N^{-1/2})$ control and **no feature-map enlargement** (contrast the level-$2M$ enlargement of Prop. 5.1); this is the estimation-theoretic face of the full-signature character (Prop. 5.3(1)). In fitting, it means the variance is a **log-link GLM** — regress $\log V$ linearly in b, or equivalently a Gamma/log-link on V — which is linear-in-parameters and keeps the fitted variance positive by construction. **What is lost** is, as with the quadratic readout, the affine transform property (Fourier/Riccati pricing $\to$ Monte-Carlo on the stored reservoir paths); the further, exponential-specific trades — the lognormal-basket VIX^2 and the leverage-sign-dependent martingale property — are taken up in the comparison below.

Log-link GLM (definition). A generalized linear model (McCullagh–Nelder) predicts a response Y from features x through three ingredients: a *linear predictor* $\eta = \theta^\top x$; a *link* g relating it to the conditional mean, here the **log link** $g = \log$, so that $\mu(x) := \mathbb{E}[Y \mid x] = \exp(\theta^\top x) > 0$; and an exponential-family *response law* (e.g. Gamma or quasi-Poisson) fixing the mean–variance relation $\operatorname{Var}(Y \mid x) = \phi\, V(\mu)$ via a variance function V and dispersion ϕ. The parameter is fit by minimising the total deviance,

$$\hat\theta = \arg\min_\theta \sum_p \operatorname{Dev}\big(Y^{(p)},\, \mu(x^{(p)})\big), \tag{5.8}$$

whose stationarity is an *unbiased estimating equation* $\sum_p x^{(p)}\big(Y^{(p)} - \mu(x^{(p)})\big) = 0$ (for the log-canonical / quasi-Poisson family; Gamma reweights the residual by $1/\mu$ but keeps the same target). Since $\mathbb{E}[Y - \mu \mid x] = 0$ at the truth, the fit targets the **arithmetic** conditional mean $\mathbb{E}[Y \mid x]$ — unlike least squares on $\log Y$, which converges to the *geometric* mean $\exp(\mathbb{E}[\log Y \mid x]) \le \mathbb{E}[Y \mid x]$ and so under-predicts a positive convex target. Here $Y = V$ (or a forward-variance sample), $x = R$ (or a random feature $\Phi(R)$), and θ carries the loading b: the log link *is* the exponential readout $V = \exp(b^\top R + \text{const})$, giving positivity for free and unbiasedness for the pricing-relevant arithmetic mean.

Quadratic vs exponential: what transfers and what trades. The two nonlinear readouts share the essential Zumbach content and differ mainly in tractability. *What transfers.* Sign-folding is automatic for the exponential — its ≥ 2 shuffle powers already contain the trend-magnitude term — and the strong-Zumbach master formula holds **verbatim**, with the effective quadratic $Q = \frac{1}{2} b b^\top$ and a clean prefactor: Corollary 9.9 gives $\mathcal{Z}^{\mathrm{exp}}_{\mathrm{strong}}(\tau) = 4\xi_0\,(b^\top \mathcal{A}_\tau u)(b^\top S_\tau u)$, area-linear and vanishing iff the reservoir is reversible, exactly as for the quadratic readout (Prop. 9.7). *What trades.* (i) The exponential's effective Q is **rank-one** ($\frac{1}{2} b b^\top$), so a single loading vector b shapes the area coupling, whereas a full symmetric Q offers $\frac{1}{2}N(N+1)$ knobs — the exponential is less flexible for *sculpting* the asymmetry but buys positivity and all higher orders for free. (ii) It **loses the affine-VIX^2 property**: under the exponential readout the forward variance is exponential-affine and VIX^2 is a correlated-lognormal *basket* rather than an affine map of the state (Prop. 9.2 and its exponential remark) — Bergomi's well-known VIX difficulty. (iii) Its **martingale property is delicate**: a lognormal V has no exponential moments, so Novikov fails and the true-martingale property of S must be argued à la Sin / Lions–Musiela / Andersen–Piterbarg (Gassiat, for rough Bergomi), with the leverage sign decisive (non-positive correlation the safe regime). By contrast the *softplus* readout $\log(1 + e^{w^\top R})$ grows only linearly, so on the Gaussian reservoir it has a Gaussian right tail and Novikov *holds* (T7): softplus is the martingale-safe positive readout, the exponential the Bergomi-realistic but delicate one. A positive readout combining both — an exponential-of-quadratic $V = \exp(b^\top R + \frac{1}{2} R^\top M R)$ — stays positive and keeps Gaussian-integral moments under $I - PM \succ 0$, at the price of that moment constraint.

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6. (b) Random → structured vector fields

CGGOT draw (A_i, b_i) **at random**; this is what makes the universality proof a random-projection/concentration argument. The exact signature is the opposite extreme — fully *structured* (canonical shift operators). Between them lies a design spectrum, and the ESN literature already exploits it.

- **Random (CGGOT, Theorem 3.6).** Maximal genericity; universality via Johnson–Lindenstrauss-type concentration; no interpretability of individual coordinates.
- **Structured / minimal (Rodan–Tiňo simple cycle reservoirs; Jaeger).** Deterministic, sparse, sometimes provably as expressive as random ones for given memory; coordinates carry identifiable roles.
- **Designed-coordinate (the proposal here).** Choose the B_i so that *specific* reservoir coordinates are *named* signature words. The case of interest: the **Lévy-area coordinates** between the price-driver Z (defined in 2.1′, correlated to W via ϱ) and the volatility-carrying channels.

Definition 6.1 (Lévy area; the time-irreversible level-2 words). For channels i, j, the **Lévy area** is the antisymmetric part of the level-2 signature,

$$\mathbb{A}^{ij}_{0,t} = \tfrac{1}{2}\big(\mathbb{X}^{ij}_{0,t} - \mathbb{X}^{ji}_{0,t}\big) = \tfrac{1}{2}\int_0^t \big(X^i_s\, dX^j_s - X^j_s\, dX^i_s\big). \tag{6.1}$$

Lemma 6.2 (area is odd under time reversal). Let $\overleftarrow{X}_s := X_{t-s} - X_t$ be the reversed path. Then $\mathbb{S}(\overleftarrow{X})_{0,t} = \mathbb{S}(X)^{-1}_{0,t}$ (the antipode in $T((\mathbb{R}^d))$), and on the level-2 antisymmetric part $\mathbb{A}^{ij} \mapsto -\mathbb{A}^{ij}$, whereas the symmetric part $\frac{1}{2}(\mathbb{X}^{ij} + \mathbb{X}^{ji}) = \frac{1}{2}X^i X^j$ is **even**. Therefore **time-reversal asymmetry of a signature-linear functional is carried precisely by its loading on the antisymmetric (area / lead–lag) words.**

Consequence. To manufacture a controlled, *interpretable* time-reversal asymmetry — the structural fingerprint of Zumbach — one **structures** a handful of reservoir coordinates to be the areas $\mathbb{A}^{Z,\text{vol-channel}}$ and loads them in the (linear or quadratic) readout. This is *more general* than either Cuchiero (random) or Abi Jaber (exact-but-linear): random where genericity/universality is wanted, structured where a named stylised fact is targeted. It also connects to the operator-algebra view: the area between channels is generated by the **commutator** $[B_i, B_j]$ in (3.1)/(3.4), so "loading area" ≡ "using non-commuting (hence non-normal) vector fields," tying §6 to the irreversibility discussion in §9.2 (and to T4).

Relation to §5(a) — these compose, they are not alternatives. The two observations act on different layers and are meant to be used together. The **quadratic readout (5a)** is a *readout-layer* device: its role is the *sign-folding* (it turns a signed trend coordinate into a trend-*magnitude* effect, $T \mapsto T^2$), and it requires no special reservoir. The **designed Lévy-area coordinates (6b)** are a *reservoir-architecture* device: their role is to *create* the time-irreversible features inside the state R in the first place, as explicit and interpretable coordinates. The link is the shuffle identity of Proposition 5.1: a quadratic readout on a level-1 reservoir already reaches level-2 words, and the *antisymmetric* part of those words is exactly the Lévy area (Lemma 6.2). Thus (5a) reaches area-type terms only *implicitly*, bundled with the symmetric level-2 terms; (6b) isolates them as separate coordinates a *linear* readout can load directly. The natural design uses both: (6b) puts $\mathbb{A}^{Z,n}$ in the state, and (5a) supplies the convex/sign-folding readout that loads them and folds the remaining trend coordinates. Theorem T3 below is phrased in terms of a single area-loading weight β precisely because that is the clean object for a theorem; operationally β is realised either by structuring coordinates (6b), by a quadratic readout (5a), or by both.

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7. (c) The random controlled differential equation (RCDE) umbrella

Definition 7.1 (RCDE reservoir). A **random controlled differential equation** reservoir is

$$dR_t = V_0(R_t)\,dt + \sum_i V_i(R_t) \circ dU_t^i, \qquad V_i \text{ random/frozen}, \qquad V_t^{\text{var}} = h(R_t), \tag{7.1}$$

driven by a **control** U. This single object specialises to everything above and provides a *continuous dial* between two regimes:

- **Exogenous control $U = \widehat{W} = (t, W, Z)$.** With W the $(d-1)$-dimensional reservoir Brownian and Z the scalar price Brownian of 2.1′ (so $dZ_t = \varrho^\top dW_t + \sqrt{1-\|\varrho\|^2}\,dB_t^\perp$, $B^\perp \perp W$ scalar), augmenting the control with Z — equivalently with the single idiosyncratic scalar $B^\perp$, one channel beyond the d of $\widehat{W}$ — is what gives the reservoir access to the price-driving direction, hence to the cross-channel Lévy-area coordinates $\mathbb{A}^{Z,n}$ of §6. With this control, R depends only on exogenous Brownians, hence is **simulated once**; the price-side parameters (readout w, Q, c_0; leverage ϱ) are pure **readout** choices re-evaluated on stored paths. This is the **Cuchiero/Abi-Jaber reusability** regime (and recovers Definition 2.8 when h is linear and $\sigma = \mathrm{id}$). It delivers the **weak** Zumbach effect and the leverage skew, but — because R is a functional of $(W, B^\perp)$, not of the realised S-path — **not** the strong effect through U alone (the leverage ϱ is the *only* price→vol channel and is structurally too weak; cf. the rough-Heston weak-but-not-strong dichotomy).
- **Price-driven control $U \ni \log S$.** R is driven by the realised return path; V becomes a true path functional and the model is genuine PDV (Guyon–Lekeufack), delivering the **strong** effect — at the cost of reusability, since changing a price-dynamics parameter changes U and forces re-simulation.

The reusability–fidelity axis (summary).

Regime	Control U	Re-sim on parameter change?	Zumbach reachable
Exact signature / linear readout (Abi Jaber–Gérard)	$\widehat{W} = (t, W)$	No (readout + ϱ only)	weak
Random signature / linear readout (CGGOT)	$\widehat{W}$	No	weak
Exogenous + quadratic readout on area words (this note)	$\widehat{W} = (t, W, Z)$	**No**	**strong-*like* (faked via area/trend readout)**
Frozen reference-vol trend coordinate	$\widehat{W}$ + stored V^{ref}	No	strong-like
Self-consistent trend coordinate	$\widehat{W}$, with T driven by $\sqrt{V}\,dZ$	Partial (self-referential)	strong (faithful)
Price-driven RCDE / PDV (Guyon–Lekeufack)	$(t, \log S)$	Yes	strong (faithful)

The **third row** is the sweet spot the three observations point to: an exogenous-driven (hence reusable) reservoir, with the strong-Zumbach *consequence* injected by a **quadratic readout loaded on the Lévy-area / trend coordinates** — fidelity to the *observable asymmetry* without literal price feedback.

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8. Extra thoughts: theorems that look provable

Below, **(K)** marks essentially-known results worth restating in this language, **(F)** feasible new results, **(O)** open/conjectural with a concrete strategy. The numbering is for reference, not priority.

T1 (K). Volterra–signature equivalence for the volatility reservoir. *Statement.* For the state-affine volatility reservoir (Definition 2.8 / 2.6), the order-M Volterra kernels of V are the exponential-resummed level-$\leq M$ signature coefficients (3.4), and the truncations coincide (3.5). *Status.* Theorem 3.2 above; the content is to state it cleanly for $\sigma_t = \langle \ell, \mathbb{S}(\widehat{W}) \rangle$ and to identify the fractional-loading kernel quadrature (Corollary 3.3). *Use.* Gives the signature-volatility model an exact ESN/Volterra reading and a dimension-explicit error notion (level $\leftrightarrow$ order $\leftrightarrow$ #factors).

T2 (F). Reusability theorem for quadratic readouts. *Statement.* Let R solve the exogenous RCDE (7.1) with $U = \widehat{W}$. For any (c_0, w, Q, ϱ) the variance (5.1) and the price S are measurable functionals of the **fixed** pair (R, Z); consequently the map $(c_0, w, Q, \varrho) \mapsto$ (model prices) factors through statistics of stored paths, and its gradient is a pathwise/AAD object requiring **no re-simulation**. Moreover the affine (Fourier/Riccati) transform holds iff $Q = 0$. *Strategy.* Pathwise measurability + dominated convergence for the gradient; the affine claim is the Abi Jaber–Gérard characteristic-functional computation, which uses linearity of h at exactly one step. *Use.* Formalises precisely which parameters are "free" (readout, ϱ) vs "frozen" (reservoir), i.e. the reusability that motivates the whole programme.

T3 (O; partial results in §9.2). Area-loading ⇒ strong Zumbach (the central conjecture). *Statement (leading order).* Consider the exogenous reservoir with a readout loading β on the Lévy-area coordinates $\mathbb{A}^{Z,n}$ between the price-driver Z and vol channels n. Then the strong-Zumbach functional — e.g. the regression coefficient of future realised variance on a **signed** past-trend feature, controlling for past realised variance — satisfies, to leading order in the vol-of-vol ν and day-length δ,

$$\mathcal{Z}_{\mathrm{strong}}(\tau) \;\approx\; \langle \beta,\; g(\tau)\rangle \cdot c(\varrho, \nu, \delta), \tag{8.1}$$

with $g(\tau)$ determined by the reservoir kernels (3.4) and c an explicit prefactor; in particular $\mathcal{Z}_{\mathrm{strong}} \equiv 0$ if $\beta = 0$ (pure symmetric/diagonal readout) and the **sign of $\mathcal{Z}_{\mathrm{strong}}$ is the sign of β along g.** *Strategy.* This is the ESN analogue of the El Euch–Gatheral–Radoičić–Rosenbaum $Z(\tau) \sim 2(\rho\nu)^2\delta^{2H+1}g_\alpha(k)\xi_0$ result: expand the estimator as a fourth-order functional of (W, Z), use Wick/Isserlis to reduce to products of the kernels (3.4), and isolate the antisymmetric (area) contribution via Lemma 6.2. *Status.* The two **structural** claims are now established in the linear-Gaussian corner with a **quadratic** readout ($Q \neq 0$; note a linear readout gives $\mathcal{Z}_{\mathrm{strong}} \equiv 0$ by odd Wick, so $Q = 0$ is *not* the right corner): Prop. 9.7 (master formula — reversible ⇒ vanishing, and exact linearity in the Lévy-area part $\mathcal{A}_\tau$, with sign) and Prop. 9.8 (a closed-form two-factor instance). What remains open is the quantitative rough scaling $\mathcal{Z} \sim \delta^{2H+1}$ (the $N \to \infty$ log-grid limit plus the short-window δ-expansion) and the transfer through a nonlinear readout. *Use.* Turns "load the area terms" from heuristic into a quantitative design rule, and is the paper's theoretical nugget.

T4 (F/O). Non-normality ↔ irreversibility ↔ area, unified. *Statement.* For the linear-Gaussian reservoir $dR = AR\,dt + \Sigma\,dW$ with stationary law $\pi = \mathcal{N}(0, P)$ $(AP + PA^\top + \Sigma\Sigma^\top = 0)$: (i) the process is time-reversible $\iff$ $A\Sigma\Sigma^\top$ is symmetric $\iff$ the stationary probability current $\mathcal{J} \equiv 0$; (ii) the magnitude of the generated Lévy areas $\mathbb{E}[{\mathbb{A}^{ij}_{0,t}}^2]$ and of $\mathcal{Z}_{\mathrm{strong}}$ are controlled by a norm of the commutator $[A, A^\top]$ (departure from normality) and of $A\Sigma\Sigma^\top - (\,\cdot\,)^\top$. *Strategy.* Closed-form: reversibility/detailed-balance for linear diffusions is classical (the current is $\mathcal{J}(r) =$ (antisymmetric part of AP) $\pi(r)/P$-type expression); the area variance is a Gaussian computation; bound both by $\|[A, A^\top]\|$. *Status.* (i) known; (ii) feasible in the Gaussian case, open in the nonlinear case. *Use.* Ties the cascade/non-normal-drift route, the area/signature route, and the physics of broken detailed balance into one statement.

T5 (F). $O(N^{-1/2})$ approximation of the strong-Zumbach kernel. *Statement.* The second-order Volterra (strong-Zumbach) kernel of any FMP volatility functional is approximated in L^2 at rate $O(N^{-1/2})$ by an N-dimensional randomized-signature reservoir with a **quadratic** readout. *Strategy.* Combine Boyd–Chua (the target is a finite Volterra series, Theorem 3.5) with the Gonon–Grigoryeva–Ortega random-feature bounds (Theorem 3.6) applied to the enlarged feature map $r \mapsto \mathrm{vech}(rr^\top)$ (Proposition 5.1). *Status.* Feasible; essentially an application of existing bounds to the quadratic feature set. *Use.* Gives the "fake strong Zumbach with a reusable ESN" approach a convergence guarantee.

T6 (K, carried out in §9.1). Linear-Gaussian benchmark (closed forms). *Statement.* For $V_t = a + b^\top R_t$ (or $+R_t^\top Q R_t$) with linear-Gaussian R, closed forms exist for $\mathbb{E}[V_t]$, $\mathrm{Cov}(V_t, V_{t+\tau})$, $\mathbb{E}[\int_0^T V]$ and its variance, the VIX map $m_T(r)$, and the leverage function $L(\tau)$. *Strategy.* Gaussian moments of OU + Wick. *Status.* **Done in §9.1** (Prop. 9.1–9.3 and (9.13)); elementary, not claimed novel. *Use.* Validates the Monte-Carlo engine, the estimators, and the pathwise gradients; and is the testbed for T3/T4.

T7 (F). ESP/true-martingale region under a quadratic readout. *Statement.* If the reservoir has bounded (e.g. tanh) diffusion so R is sub-Gaussian, and the readout (5.1) has $Q \succeq 0$ with at-most-quadratic growth controlled by $\|Q\|$, then ESP holds and $S = \mathcal{E}(\int \sqrt{V}\,dZ)$ is a true $\mathbb{Q}$-

martingale on $[0,T]$ for $\|Q\|$ below an explicit bound (and any leverage with non-positive projection on the vol-of-vol direction). *Strategy.* Lyapunov/Meyn–Tweedie drift condition for ESP; Novikov via sub-Gaussian moments for the martingale property; mirror the Abi Jaber–Gérard "martingality depends on truncation/parameters" analysis for the quadratic case. *Status.* Feasible. *Use.* Keeps the faked-strong-Zumbach model arbitrage-free, and identifies the admissible region for (w, Q, ϱ). *Exponential / softplus readouts.* On the Gaussian reservoir it is the *tail of the readout* that decides. The **softplus** $V = \log(1 + e^{w^\top R})$ grows only linearly, so V has a Gaussian right tail and the same Novikov argument applies — it is the martingale-safe positive readout. The **exponential** $V = \xi_0 e^{b^\top R - \frac{1}{2} b^\top P b}$ is lognormal and has **no exponential moments**, so Novikov fails; the true-martingale property then needs the Sin / Lions–Musiela / Andersen–Piterbarg analysis (Gassiat, for rough Bergomi) and hinges on the leverage sign (non-positive projection on the vol-of-vol direction being the safe regime). This is a *readout-level* tail effect, distinct from the *diffusion-level* heavy tails that arise when the nonlinearity sits in $\Sigma(R)$ instead (§3.4(i)).

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9. Worked results: T6 in closed form, and partial results toward T3

> **Authorship flag (read first).** Everything in this section is derived by the author for this note; none of it is quoted from a source. The status of each piece, however, differs and is stated explicitly: - **§9.1 (T6).** Elementary Gaussian / Ornstein–Uhlenbeck computations. Standard *technique* (Wick's theorem, Lyapunov equations); assembled here as a benchmark. **Not claimed as novel** — any of these could be reconstructed by a competent reader, and close relatives appear in the affine/Bergomi and polynomial-diffusion literatures. - **§9.2, Prop. 9.4–9.5.** To the author's knowledge these exact statements — the zero-leverage vanishing for an *arbitrary* readout, and the leverage-vs-irreversibility source decomposition in *reservoir* form — are **not written down in the literature in this form**. The underlying mechanism is implicit in El Euch–Gatheral–Radoičić–Rosenbaum (EGRR, 2018); the contribution here is the clean reservoir-level statement and proof. They are elementary given the setup, so the honest label is *"new framing of folklore, rigorously stated."* - **§9.2, Prop. 9.7–9.8 (strong-Zumbach master formula and its closed-form instance).** Same status and same honest label as 9.4–9.5: elementary Gaussian algebra (Wick + Lyapunov + matrix exponential), symbolically machine-checked, not previously written in this reservoir form to the author's knowledge. They establish the *structural* content of Conjecture 9.6 (reversible-vanishing; exact area-linearity with sign) in the linear-Gaussian, quadratic-readout corner. - **Conjecture 9.6 (= T3).** The *structural* claims are now proved (Prop. 9.7–9.8); what remains genuinely **open** is the quantitative rough scaling $\mathcal{Z} \sim \delta^{2H+1}$ (the $N \to \infty$ / short-window limit) and the transfer through a nonlinear readout. Where a constant would require an expansion the author has not verified to be internally consistent, **no constant is stated** — only the robust structural claims. This is deliberate.

The logic of this section (roadmap). The two subsections are organised by one principle, dictated by what we are trying to measure.

- **§9.1 collects, in closed form, every *two-point* (second-order) statistic of the benchmark:** the variance autocovariance, the VIX conditional expectation, the quadratic-readout

autocovariance, and the return–variance leverage function.
- **§9.2 is what happens at the first order where an *arrow of time* can appear:** the Zumbach asymmetry.

The split is forced, not cosmetic. A stationary *scalar* autocovariance is automatically time-symmetric, so nothing in §9.1 can see an arrow of time; the Zumbach effect, sign-blind by construction, is therefore necessarily a *four-point* object and must live in §9.2. The connective tissue throughout is **two matrices**: the stationary covariance P (from the Lyapunov equation) and the propagator $e^{A\tau}$ (from the Markov/semigroup property). Solve for P once and propagate by $e^{A\tau}$, and every second-order statistic falls out; push the same calculation to fourth order and the time-asymmetry diagnostics appear, sourced by exactly two mechanisms — leverage and reservoir non-normality. The reader can hold onto this single arc: *one linear solve (P) + one matrix exponential ($e^{A\tau}$)   all of §9.1; the same machinery at fourth order   §9.2.*

Terminology: k-point functions. We use the physicists' name — standard in the econophysics / Zumbach literature this section connects to (Bouchaud and co-authors; EGRR) — though it is only semi-standard in statistics, so we fix it once. A **k-point function** of a process X is an expectation of a product of k values of the process at k times,

$$\mathbb{E}\big[X_{t_1} X_{t_2} \cdots X_{t_k}\big], \tag{9.1}$$

the "point" being a time argument and k counting the factors inside the expectation. A **two-point function** is therefore, after centering, the autocovariance $\mathrm{Cov}(X_s, X_t) = \mathbb{E}[X_s X_t]$ — i.e. *second-order* structure — and a **four-point function** is $\mathbb{E}[X_{t_1} \cdots X_{t_4}]$. In mainstream statistics the same content goes by *second-order structure* (the two-point case) and, for the part of a four-point function that does **not** reduce to products of two-point functions, the *connected four-point function* or *fourth-order cumulant*. The count is what organises §9: by the Wick–Isserlis identity (Lemma 9.0) a Gaussian process's two-point function determines all its higher-point functions, so the second-order theory of §9.1 is closed by P and $e^{A\tau}$ alone, whereas a genuine arrow of time first appears in the connected four-point function of §9.2. (Strictly, two of the §9.1 objects — the VIX conditional mean and the integrated-variance mean — are *first*-order, so "second-order structure" is the precise umbrella for §9.1; we keep "two-point" for the autocovariances, where it is exact.)

9.1 The linear-Gaussian benchmark (T6)

Why this benchmark. Before trusting the full nonlinear, state-dependent reservoir in Monte Carlo, we want a corner of the model where *everything is computable by hand*, to serve as ground truth for the simulation engine, the VIX regression, and the pathwise-gradient code. Freezing the diffusion $\Sigma(R) \equiv \Sigma$ and keeping a linear drift makes R an Ornstein–Uhlenbeck process — Gaussian, hence completely characterised by its first two moments — which is exactly such a corner (the Gaussian/Bergomi corner of the family):

$$dR_t = AR_t\,dt + \Sigma\,dW_t, \qquad A \in \mathbb{R}^{N\times N} \text{ stable}, \quad \Sigma \in \mathbb{R}^{N\times m},\ m = d-1. \tag{9.2}$$

Why the Lyapunov equation. The reservoir covariance $P_t = \mathrm{Cov}(R_t)$ obeys the deterministic matrix ODE $\dot P_t = AP_t + P_t A^\top + \Sigma\Sigma^\top$ — the two drift terms come from $d(R_t R_t^\top)$, the $\Sigma\Sigma^\top$ from the quadratic variation. Its stationary value is the fixed point $\dot P = 0$, i.e. the **algebraic Lyapunov equation**

$$AP + PA^\top + \Sigma\Sigma^\top = 0, \tag{9.3}$$

with unique positive-definite solution $P = \int_0^\infty e^{As}\Sigma\Sigma^\top e^{A^\top s}\,ds$ whenever A is Hurwitz (spectrum in the open left half-plane). We foreground it because it does **three jobs at once**:

1. *It is the single sufficient statistic for the whole benchmark.* Every formula below is built from P and the matrix exponential $e^{A\tau}$; one $N \times N$ linear solve therefore closes the entire second-order theory. This is the "compute the reservoir statistics once" spirit of the reusability programme made literal.
2. *Its solvability is the Echo State Property.* A unique positive-definite P exists iff A is stable, i.e. iff the reservoir has a well-defined stationary (fading-memory) regime. So (9.3) is simultaneously the ESP/stability certificate that the Pallavicini note only *assumes* (its §3.5 gap) — here it is an algebraic solvability condition.
3. *It guarantees stationarity.* The stylised facts (variance autocorrelation, Zumbach, SSR) are time-translation-invariant objects and only make sense in a stationary regime; the Lyapunov solution is what puts us there rather than in a transient.

Why $C_\tau = Pe^{A^\top \tau}$. By the Markov property, $R_{t+\tau} = e^{A\tau} R_t + \int_t^{t+\tau} e^{A(t+\tau-s)} \Sigma \, dW_s$, whose stochastic-integral term is mean-zero and independent of $\mathcal{F}_t$; hence the lagged covariance is the static covariance *propagated forward by the semigroup* $e^{A\tau}$:

$$C_\tau := \mathrm{Cov}(R_t, R_{t+\tau}) = P \, e^{A^\top \tau} \qquad (\tau \geq 0). \tag{9.4}$$

This is the second load-bearing object: *all* time-lagged structure is P filtered through $e^{A\tau}$. Consequently every result in §9.1 has the same shape — a readout's autocovariance is "P propagated by $e^{A\tau}$, contracted with the readout weights" — and the reader can predict the form of each before seeing it.

Positivity caveat. A linear readout $V = a + b^\top R$ is not almost-surely positive; this corner is meant for validating moments, the VIX map, the estimators, and the pathwise gradients — **not** for production pricing. The quadratic readout (below) or a capped/exponential readout restores positivity. We write $\varsigma := \Sigma^\top b \in \mathbb{R}^m$ for the **vol-of-vol loading** (reserving β for the area-loading weight of T3, with which it must not be confused).

Lemma 9.0 (Wick–Isserlis). *Let $(G_1, \ldots, G_{2n})$ be jointly Gaussian and mean-zero. Then every higher moment reduces to a sum, over all pairings of the indices, of products of covariances:*

$$\mathbb{E}\big[G_1 G_2 \cdots G_{2n}\big] = \sum_{\text{pairings } \mathcal{P}} \prod_{\{i,j\} \in \mathcal{P}} \mathbb{E}[G_i G_j], \tag{9.5}$$

the sum running over the $(2n-1)!!$ partitions of $\{1, \ldots, 2n\}$ into n unordered pairs; every odd moment vanishes. In particular

$$\mathbb{E}[G_1 G_2 G_3 G_4] = \mathbb{E}[G_1 G_2] \, \mathbb{E}[G_3 G_4] + \mathbb{E}[G_1 G_3] \, \mathbb{E}[G_2 G_4] + \mathbb{E}[G_1 G_4] \, \mathbb{E}[G_2 G_3]. \tag{9.6}$$

This is the only computational engine §9 needs. It is what makes the linear-Gaussian corner *closed-form* — every moment of V is a polynomial in the entries of P and $e^{A\tau}$ — and it is precisely what "reduce to products of the kernels" means in the fourth-order arguments of §9.2 and Conjecture 9.6. Two immediate consequences used below: the odd-moment vanishing kills the linear×quadratic cross-covariance in Prop. 9.3 and drives the parity arguments behind the leverage function (9.13) and Prop. 9.4; and the two *connected* (non-disconnected) pairings of $\mathbb{E}[(R_t^\top Q R_t)(R_{t+\tau}^\top Q R_{t+\tau})]$ are what produce the $2 \, \mathrm{tr}(Q C_\tau Q C_\tau^\top)$ term of Prop. 9.3.

Proposition 9.1 (linear readout; second-order structure). *For $V_t = a + b^\top R_t$ at stationarity,*

$$\mathbb{E}[V_t] = a, \qquad \mathrm{Var}(V_t) = b^\top P \, b, \qquad \mathrm{Cov}(V_t, V_{t+\tau}) = b^\top P \, e^{A^\top \tau} b \;\; (\tau \geq 0). \tag{9.7}$$

If $A = -\operatorname{diag}(\kappa_1, \ldots, \kappa_N)$ and $P = \operatorname{diag}(p_n)$ then $\operatorname{Cov}(V_t, V_{t+\tau}) = \sum_n b_n^2 p_n e^{-\kappa_n \tau}$ — a completely monotone, positive-definite autocovariance that approximates a power law across $[\kappa_{\max}^{-1}, \kappa_{\min}^{-1}]$, i.e. the multi-factor Markovian proxy of a rough kernel (Corollary 3.3).

Why we state it / where used. This is the closed-form testbed for the central roughness claim of Corollary 3.3: the sum of exponentials $\sum_n b_n^2 p_n e^{-\kappa_n \tau}$ is exactly the multi-factor proxy that must imitate a power law τ^{2H} over the band $[\kappa_{\max}^{-1}, \kappa_{\min}^{-1}]$ — one sees here precisely how the imitation works over a finite range and where it must break (the *pre-asymptotic* roughness story). It is also the analytic target against which the simulated variance autocorrelation is checked.

Proposition 9.2 (integrated variance and the VIX map). *The forward integrated-variance conditional expectation is affine in the state:*

$$m_T(r) := \mathbb{E}\left[\frac{1}{\tau}\int_T^{T+\tau} V_s\, ds \,\middle|\, R_T = r\right] = a + b^\top\Big(\tfrac{1}{\tau} A^{-1}\big(e^{A\tau} - I\big)\Big)\, r, \tag{9.8}$$

the rigorous Markov form of the VIX^2 conditional expectation. Its unconditional mean is a, and the variance of integrated variance is

$$\operatorname{Var}\Big(\int_0^T V_s\, ds\Big) = 2\int_0^T (T-\tau)\, b^\top P\, e^{A^\top \tau} b\, d\tau. \tag{9.9}$$

Why we state it / where used. VIX^2 is a conditional expectation of forward integrated variance, and in the benchmark it is a *known affine map* of the state — so it is ground truth for unit-testing the VIX regression (KC/RN, or any), including its bias, where otherwise one only has noisy Monte-Carlo targets. The variance-of-integrated-variance is the convexity quantity that separates the variance-swap level from the VIX (the Jensen gap), hence the benchmark for the convexity the estimator must reproduce.

Under an exponential readout. With $V_t = \xi_0 \exp\big(b^\top R_t - \frac{1}{2} b^\top P b\big)$ (§5, Prop. 5.3) the forward variance is instead **exponential-affine** in the state,

$$\mathbb{E}_T[V_s] = \xi_0\, \exp\Big(-\tfrac{1}{2} b^\top P b + \tfrac{1}{2}\, q(s-T) + b^\top e^{A(s-T)} R_T\Big), \qquad q(v) := b^\top \int_0^v e^{Aw} \Sigma\Sigma^\top e^{A^\top w}\, dw\, b, \tag{9.10}$$

so $\mathrm{VIX}^2 = \frac{1}{\tau}\int_T^{T+\tau} \mathbb{E}_T[V_s]\, ds$ is a **correlated-lognormal basket** rather than an affine map of R_T — the Bergomi forward-variance-curve structure. It is still fully specified (the $\mathbb{E}_T[V_s]$ are log-linear in R_T), but the clean affine $m_T(r)$ above is replaced by the lognormal-basket VIX that Bergomi models handle by moment-matching. This is the affine-VIX^2 tractability that the exponential readout trades away (§5, "what trades").

Proposition 9.3 (quadratic readout; second-order structure). *For $V_t = a + b^\top R_t + R_t^\top Q R_t$, $Q = Q^\top$, at stationarity,*

$$\mathbb{E}[V_t] = a + \operatorname{tr}(QP), \tag{9.11}$$

$$\operatorname{Cov}(V_t, V_{t+\tau}) = \underbrace{b^\top P e^{A^\top \tau} b}_{\text{linear part}} + \underbrace{2\, \operatorname{tr}\big(Q\, C_\tau\, Q\, C_\tau^\top\big)}_{\text{quadratic part}}, \qquad C_\tau = P e^{A^\top \tau}. \tag{9.12}$$

Proof. Lemma 9.0 applied to the zero-mean jointly Gaussian $(R_t, R_{t+\tau})$: the linear×quadratic cross-covariance vanishes (odd third moment), and $\operatorname{Cov}(R_t^\top Q R_t,\, R_{t+\tau}^\top Q R_{t+\tau}) = 2\operatorname{tr}(Q C_\tau Q C_\tau^\top)$ from the two connected pairings (the disconnected pairing reproduces $\mathbb{E}[R_t^\top Q R_t]\, \mathbb{E}[R_{t+\tau}^\top Q R_{t+\tau}]$ and is removed by the covariance). ☐

Structural reading. The quadratic part is **bilinear in C_τ**, hence decays at **twice the rate** of the linear part — an exact, testable fingerprint of a Q-readout in the variance autocorrelation, and the second-order-Volterra (T^2) channel of §5(a) made quantitative. *Where used:* this is both an implementation check (a genuine Q-readout must show two decay scales in the variance autocorrelation) and an *identifiability* handle — b and Q are separable by their decay rates, which is what keeps the joint calibration of (w, Q) well-posed rather than degenerate.

Under an exponential readout. The lognormal analogue (Prop. 5.3) is $\mathrm{Cov}(V_t, V_{t+\tau}) = \xi_0^2\big(e^{c(\tau)} - 1\big)$ with $c(\tau) = b^\top P e^{A^\top \tau} b$: a *single* decay object $c(\tau)$ — the log-variance autocovariance, the memory kernel of the variance — exponentiated, in place of the two-rate linear-plus-quadratic split above; for small c it reduces to $\xi_0^2\, c(\tau)$, matching the linear part. The leverage follows from $\mathrm{Cov}(e^G, H) = e^{\frac{1}{2}\mathrm{Var}\, G}\,\mathrm{Cov}(G, H)$ in the same way (9.13) uses the linear covariance. (The memory structure across all three readouts is collected in §9.4.)

Leverage function (return level). With $dZ_t = \varrho^\top dW_t + \sqrt{1 - \|\varrho\|^2}\, dB_t^\perp$ and the driftless short-window return $r_t = \int_t^{t+\delta} \sqrt{V_s}\, dZ_s$, a first-order expansion in δ (linear readout) gives

$$L(\tau) := \mathrm{Cov}(r_t,\, V_{t+\tau}) = \delta\, \mathbb{E}[\sqrt{V_t}]\; b^\top e^{A\tau} \Sigma \varrho \quad (\tau > 0), \qquad L(\tau) = 0 \;(\tau < 0). \tag{9.13}$$

This is the skew/leverage object: it is nonzero iff ϱ has a component along $\Sigma^\top e^{A^\top \tau} b$, decays through the reservoir spectrum, and defines the **instantaneous price–variance covariation rate** $\rho_{ZV} := b^\top \Sigma \varrho = \varsigma^\top \varrho$. (The one-sided support in τ — future variance responds to a past return, not conversely — is the signed precursor of the Zumbach asymmetry; the *even* version requires the fourth-order computation of §9.2.)

Why we state it / where used. It is the model's skew engine — the calibrated ρ_{ZV} is what tilts the ATM smile — and it is the *signed, two-point warm-up* for the sign-blind, four-point Zumbach object of §9.2: same price→variance channel, one order simpler. Its one-sidedness already exhibits, in the simplest possible form, the time-directedness that §9.2 must capture for *magnitudes* rather than signs.

9.2 Partial results toward T3 (the Zumbach asymmetry)

The hinge from §9.1: why Zumbach must be a four-point object. This is the rationale that justifies treating §9.2 separately at all. A stationary *scalar* process has an automatically **even** autocovariance: $b^\top C_\tau b$ is a scalar, so it equals its own transpose $b^\top C_\tau^\top b = b^\top C_{-\tau} b$. The variance autocorrelation of §9.1 *therefore cannot encode an arrow of time.* What *can* be asymmetric at the two-point level is the **cross** covariance between returns and variance — and indeed the leverage function (9.13) is one-sided in τ — but that is the *signed* (leverage) feedback. The Zumbach effect is by design **sign-blind**: it asks whether trend *magnitude* predicts future volatility more than the converse, so it is built from squared returns / realised variance and is intrinsically a **four-point** function. Four-point is thus the *lowest* order at which a sign-blind arrow of time can appear, and §9.2 is precisely "§9.1 pushed to that order." The signed two-point object $L(\tau)$ was the warm-up; the sign-blind four-point object $Z(\tau)$ is the real target.

Throughout use the driftless short-window conventions $r_u = \int_u^{u+\delta} \sqrt{V_s}\, dZ_s$, $RV_u = \int_u^{u+\delta} V_s\, ds$, and the continuous-time *definition-1* estimator

$$Z(\tau) = \mathrm{Cov}\big(RV_{t+\tau},\, r_t^2\big) - \mathrm{Cov}\big(RV_t,\, r_{t+\tau}^2\big), \qquad \tau > 0. \tag{9.14}$$

The two propositions answer, in order, the two questions a reader should ask about any candidate asymmetry: **when is it exactly zero** (Prop. 9.4 — *necessity* of a price→variance channel), and **given that it is not, what sources it** (Prop. 9.5 — *which* channels, and what each contributes).

Proposition 9.4 (zero coupling ⇒ no weak Zumbach; any readout). *Suppose the reservoir is* ***autonomous and driven by*** W ***only****, and the price driver is independent of the* *reservoir noise — precisely: $\varrho = 0$ **and** Z is not itself a reservoir input channel, so that $\mathcal{F}^W \perp Z$. Then for any measurable readout $V = h(R)$,*

$$Z(\tau) \equiv 0 \quad \text{for all } \tau. \tag{9.15}$$

Proof. Under the hypothesis, $Z \perp W$, so conditionally on the whole variance path $\mathcal{F}^W_\infty$ the return over any window is a time-changed Brownian motion and, by Itô isometry, $\mathbb{E}[r_u^2 \mid \mathcal{F}^W_\infty] = \int_u^{u+\delta} V_s\, ds = RV_u$ (the idiosyncratic $dB^\perp$ carries the entire second moment). Both $RV_{t+\tau}$ and RV_t are $\mathcal{F}^W$-measurable, so the tower property gives $\mathbb{E}[RV_{t+\tau} r_t^2] = \mathbb{E}[RV_{t+\tau} RV_t] = \mathbb{E}[RV_t r_{t+\tau}^2]$, and the products of marginal means coincide by stationarity. □

Interpretation. An autonomous, W-driven reservoir with $\varrho = 0$ makes **price and variance independent processes**: the variance literally cannot see the price, so no Zumbach (weak or strong) is possible. Any Zumbach signal in such a model is therefore **entirely leverage-induced** — the rigorous form of the empirical observation that a diagonal/symmetric reservoir picks up a Zumbach signal *only* through the leverage channel. Note the hypothesis is exactly what fails in the §7 construction where Z is *added to the reservoir control*: there the variance is a functional of the price-driving direction, the proposition does not apply, and a strong effect can appear (Conjecture 9.6).

Time-reversibility: the precise notion. Since "(ir)reversible" does real work in Prop. 9.5, here is exactly what it means. A stationary process $(R_t)_{t\in\mathbb{R}}$ is **time-reversible** if its law is invariant under reversing the direction of time: for every n and all $t_1 < \cdots < t_n$,

$$(R_{t_1}, R_{t_2}, \ldots, R_{t_n}) \overset{d}{=} (R_{t_n}, \ldots, R_{t_2}, R_{t_1}), \tag{9.16}$$

equivalently $(R_t)_t \overset{d}{=} (R_{-t})_t$ in the stationary regime — *a recording of the process played backwards is statistically indistinguishable from the forward one.* **Irreversible** is the negation: some finite-dimensional functional has a different law forwards than backwards, i.e. the joint law carries a detectable arrow of time. The Zumbach statistic $Z(\tau)$ is exactly such a functional — engineered to be sign-blind precisely so that it responds to a genuine arrow of time and not to the sign of leverage.

For a **Markov** process with invariant law π the notion has the standard equivalent forms: reversibility $\iff$ *detailed balance* $\pi(dx)\, p_t(x, dy) = \pi(dy)\, p_t(y, dx)$ $\iff$ the generator is **self-adjoint in $L^2(\pi)$** $\iff$ the stationary probability **current vanishes**. For the linear-Gaussian reservoir $dR = AR\,dt + \Sigma\, dW$ with stationary covariance P (so $AP + PA^\top + \Sigma\Sigma^\top = 0$), all of these reduce to the algebraic condition

$$AP = PA^\top \quad \iff \quad A\,\Sigma\Sigma^\top = \Sigma\Sigma^\top A^\top \quad \iff \quad AP = -\tfrac{1}{2}\,\Sigma\Sigma^\top, \tag{9.17}$$

which for isotropic noise $\Sigma\Sigma^\top = \sigma^2 I$ is simply $A = A^\top$ (symmetric). Two points are worth stressing because they are easy to get wrong:

- *Reversibility is a joint property of the pair $(A, \Sigma\Sigma^\top)$, not of A alone.* With isotropic — or generic, non-compensating — noise, any non-symmetric (in particular any non-normal) drift is irreversible; but a finely tuned anisotropic noise can restore detailed balance even for a non-normal A. So "non-normal ⇒ irreversible" is the *generic* statement, **exact** under isotropic noise, and it is what holds for the cascade architecture (whose fractional loading is not the compensating tuning).
- *What Prop. 9.5 actually needs is reversibility of the variance path $V_t = h(R_t)$, which R controls one-sidedly.* If R is reversible then $V = h(R)$ is reversible and $Z^{\text{irr}} = 0$; conversely

$Z^{\mathrm{irr}} \neq 0$ forces R irreversible. Irreversibility of R is thus *necessary*, and *generically sufficient*, for V to be irreversible — a degenerate readout seeing only the reversible marginals could in principle wash it out.

This is where non-normality enters; we fix it next, and relate the two.

Non-normality: definition, and the commutator / Lie-bracket link. Proposition 9.5 turns on the reservoir being *non-normal*, a term we now fix, since it is also the design target of §6 and §9.3.

Definition. A square matrix M is **normal** if it commutes with its transpose, $MM^\top = M^\top M$ — equivalently, if it is orthogonally diagonalisable (symmetric, skew-symmetric, and rotation–scaling matrices are all normal). It is **non-normal** if the **self-commutator**

$$[M, M^\top] := MM^\top - M^\top M \;\neq\; 0, \tag{9.18}$$

the size of which is measured scalar-wise by Henrici's *departure from normality* $\mathrm{dep}(M) = \big(\|M\|_F^2 - \sum_i |\lambda_i(M)|^2\big)^{1/2}$ ($= 0$ iff M is normal). A **non-normal reservoir** is one whose drift A (and/or interaction matrices B_i) is non-normal: the sparse Erdős–Rényi (Blockwise) and upper-bidiagonal (Cascade) drifts qualify, while a diagonal $A = -\operatorname{diag}(\kappa)$ does not.

Two consequences, two commutators. Non-normality enters this note through two genuinely distinct failures of commutativity, worth separating:

1. **The self-commutator $[A, A^\top]$ — irreversibility and transient growth.** The adjoint $A^\top$ governs the adjoint (formally time-reversed) evolution $e^{A^\top t} = (e^{At})^\top$, so $[A, A^\top] \neq 0$ says the forward generator and its adjoint do not commute — and equivalently that A has non-orthogonal eigenvectors. The probabilistic consequence is exactly Prop. 9.5: reversibility of the autonomous OU is the *joint* condition $A\,\Sigma\Sigma^\top = \Sigma\Sigma^\top A^\top$ on (drift, diffusion) above, which under isotropic or generic noise is $A = A^\top$; a non-normal (hence non-symmetric) drift then fails it and switches the Z^{irr} channel on. (Non-normality is strictly stronger than the minimal requirement: a *normal but non-symmetric* rotation–scaling block — the "Rough ESN" — is already irreversible. What non-normality adds is *transient amplification* of $\|e^{At}\|$, governed by the numerical abscissa / pseudospectrum rather than the eigenvalues — a stability subtlety (pseudospectral, not spectral) separate from the reversibility question here.)

2. **The cross-commutators $[B_i, B_j]$ — coupling to the Lévy area.** In the controlled equation $dR_t = \sum_i B_i R_t \, dX_t^i$ the channels act through the *linear vector fields* $V_i(r) = B_i r$, whose **Lie bracket** is the matrix commutator,

$$[V_i, V_j](r) = (DV_j)\, V_i - (DV_i)\, V_j = (B_j B_i - B_i B_j)\, r = -[B_i, B_j]\, r. \tag{9.19}$$

By the Chen–Strichartz expansion the **Lévy area** $\mathbb{A}_{0,t}^{ij} = \frac{1}{2}\int (X^i \, dX^j - X^j \, dX^i)$ enters the solution *only* through this bracket — the symmetric part of the level-2 iterated integral is the shuffle $\frac{1}{2} X_t^i X_t^j$ and carries no area. Consequently **if the B_i commute the area drops out entirely** and R sees only the increments X_t^i (the time-symmetric, level-1 information); non-commuting B_i are exactly what let R *feel* the area. Since the area is the time-reversal-*odd* part of the signature (Lemma 6.2), non-commuting interaction matrices are the algebraic prerequisite for carrying Zumbach asymmetry; the time-vs-noise area $\mathbb{A}^{0i}$ enters analogously through $[A, B_i]$.

The unifying picture. Both are instances of the same slogan — *non-commuting operators carry an oriented, time-irreversible object.* For the **autonomous** (zero-leverage) channel of Prop. 9.5

the relevant failure is $A \neq A^\top$ (the self-commutator); for the **driving-path** (area-loading) channel of §6 and Conjecture 9.6 it is $[B_i, B_j] \neq 0$ (and $[A, B_i] \neq 0$). A *diagonal* reservoir kills both at once — A is symmetric *and* the B_i are simultaneously diagonalisable, hence mutually commuting — which is precisely why it is the one architecture that can produce *no* Zumbach effect, and why the design target throughout is a genuinely non-normal, non-commuting reservoir.

Proposition 9.5 (source decomposition; the role of non-normality). *Reinstating the return drift $-\frac{1}{2}V\,dt$, at $\varrho = 0$ (autonomous W-driven reservoir) the martingale-part contribution to Z still cancels exactly, leaving a pure functional of the variance path:*

$$Z(\tau)\big|_{\varrho=0} = \tfrac{1}{4}\Big(\operatorname{Cov}\big(RV_{t+\tau},\, \Xi_t\big) - \operatorname{Cov}\big(RV_t,\, \Xi_{t+\tau}\big)\Big), \qquad \Xi_u := \Big(\int_u^{u+\delta} V_s\,ds\Big)^2. \tag{9.20}$$

*This third-order variance-path functional vanishes **iff the stationary variance path is time-reversible** (in the precise sense fixed above). For the linear-Gaussian reservoir, reversibility is the joint condition $A\Sigma\Sigma^\top = \Sigma\Sigma^\top A^\top$ (equivalently $AP = PA^\top$; detailed balance / zero stationary current), i.e. $A = A^\top$ under isotropic noise. Since reversibility is a property of the pair $(A, \Sigma\Sigma^\top)$, the clean verdict is generic rather than absolute: with isotropic — or generic, non-compensating — noise every non-symmetric, in particular every **non-normal**, drift is irreversible, so the cascade drift activates this channel (its fractional loading is not the fine tuning that would restore detailed balance). Hence, generically, even at zero leverage a non-normal reservoir produces a Zumbach asymmetry sourced **purely by the irreversibility of its own dynamics**.*

Putting the two channels together,

$$Z(\tau) = Z^{\text{lev}}(\tau) + Z^{\text{irr}}(\tau), \tag{9.21}$$

where Z^{lev} vanishes at $\varrho = 0$ and carries the EGRR leverage mechanism (their $Z \sim 2(\rho\nu)^2\delta^{2H+1}g_\alpha(k)\xi_0$), while Z^{irr} vanishes for reversible/normal reservoirs and carries the non-normality mechanism. Both are sub-leading in the short-window scaling.

Why we state it / where used. Prop. 9.5 is the rigorous seed of the slogan "use a non-normal reservoir for Zumbach": it converts a property of the *operator algebra* of A (its departure from normality, measured by the commutator $[A, A^\top]$) into a *measurable* stylised fact, and it is the bridge to T4 and to the whole structured-reservoir programme — the same non-normality that buys expressivity (Cirone et al.) is what opens the zero-leverage Zumbach channel here. It also dictates a clean numerical test of the model: fix $\varrho = 0$ and compare a diagonal (reversible) reservoir against a cascade (non-normal) one; the prediction is $Z \equiv 0$ for the former and $Z \neq 0$ for the latter, isolating the irreversibility channel from the leverage channel.

> **Flag on constants.** The proof of the $\varrho = 0$ vanishing (Prop. 9.4) and the decomposition structure (Prop. 9.5) are robust and are the author's. The **exact closed-form constants and the precise common δ-order** of $Z^{\text{lev}}, Z^{\text{irr}}$ require a fully consistent fourth-order short-window expansion that the author has **not** verified end-to-end; preliminary hand-expansions were mutually inconsistent at the constant level, so no constant is asserted here. Only the structural statements — the two vanishing conditions and the two-source split — are claimed. Closing these constants (against the linear-Gaussian forms of §9.1) is the natural next milestone and would also settle the comparison with EGRR's δ^{2H+1}.

Conjecture 9.6/T3 (area-loading ⇒ strong Zumbach; open). *Let the reservoir be driven by the **augmented exogenous control** (t, W, Z) of §7 — so that the price direction Z enters the* *state and the cross-channel Lévy-area coordinates $\mathbb{A}^{Z,n}$ (between Z and the volatility channels n) are present — and let the readout place weight β on those area coordinates. Then the **strong-Zumbach functional** (the regression coefficient of future realised variance on a **signed** past-trend feature, controlling for past realised variance) satisfies, to leading order in vol-of-vol and day-length,*

$$\mathcal{Z}_{\text{strong}}(\tau) \;\approx\; \langle \beta,\, g(\tau)\rangle\; c(\varrho, \nu, \delta), \tag{9.22}$$

with $g(\tau)$ fixed by the reservoir kernels (3.4) *and c an explicit prefactor; in particular $\mathcal{Z}_{\text{strong}} \equiv 0$ when $\beta = 0$, and its sign is the sign of β along g. The consistency requirement with Prop. 9.4 is built in: the effect needs Z in the control (else price and variance decouple and $\mathcal{Z}_{\text{strong}} = 0$).* **Status: open.** *Strategy: fourth-order Wick reduction of the estimator into products of the kernels* (3.4)*, isolating the antisymmetric contribution through the antipode/Lemma 6.2; the linear-Gaussian corner of §9.1 is the closed-form testbed.*

Progress on Conjecture 9.6: the linear-Gaussian corner in closed form. The conjecture bundles three logically separate claims — *(C1)* vanishing without the area, *(C2)* linearity in the area with a definite sign, and *(C3)* the quantitative rough-volatility scaling $\mathcal{Z} \sim \delta^{2H+1}$ of EGRR. The first two are *structural* and can be settled exactly in the benchmark of §9.1; the third is a genuine asymptotics problem (below). The next two results carry out the milestone named in T3.

Proposition 9.7 (strong-Zumbach master formula). *Let R be the stationary linear-Gaussian reservoir of §9.1 with the price direction Z among its driving channels, so that a trend coordinate $T_t = u^\top R_t$ is a functional of the past price path (the regime in which Prop. 9.4 does not force vanishing). For the quadratic variance readout $V_t = a + b^\top R_t + R_t^\top Q R_t$ and the signed past-trend feature $T_t = u^\top R_t$, define the strong-Zumbach diagnostic*

$$\mathcal{Z}_{\text{strong}}(\tau) := \operatorname{Cov}\big(V_{t+\tau},\, T_t^2\big) - \operatorname{Cov}\big(V_t,\, T_{t+\tau}^2\big), \qquad \tau > 0. \tag{9.23}$$

Write $D_\tau := \operatorname{Cov}(R_{t+\tau}, R_t) = C_\tau^\top = e^{A\tau}P$ and split it into symmetric and antisymmetric parts,

$$S_\tau := \tfrac{1}{2}\big(D_\tau + D_\tau^\top\big) = \tfrac{1}{2}\big(e^{A\tau}P + Pe^{A^\top\tau}\big), \qquad \mathcal{A}_\tau := \tfrac{1}{2}\big(D_\tau - D_\tau^\top\big) = \tfrac{1}{2}\big(e^{A\tau}P - Pe^{A^\top\tau}\big), \tag{9.24}$$

so that $D_\tau = S_\tau + \mathcal{A}_\tau$. Then

$$\mathcal{Z}_{\text{strong}}(\tau) = 2\,u^\top\big(D_\tau^\top Q D_\tau - D_\tau Q D_\tau^\top\big)u = 4\,u^\top\big(S_\tau Q \mathcal{A}_\tau - \mathcal{A}_\tau Q S_\tau\big)u. \tag{9.25}$$

In particular:

*(i) **(vanishing / reversibility)** if the reservoir is time-reversible ($AP = PA^\top$, equivalently $\mathcal{A}_\tau \equiv 0$) then $\mathcal{Z}_{\text{strong}}(\tau) = 0$ for every readout (b, Q), trend u, and lag τ; the linear part $b^\top R$ never contributes (odd third moment, Lemma 9.0).*

*(ii) **(area-linearity)** $\mathcal{Z}_{\text{strong}}$ is exactly linear in the antisymmetric part $\mathcal{A}_\tau$, whose leading behaviour is set by the reservoir's expected Lévy area. With $\mathbb{A}^{mn}_{[t,t+\tau]} = \frac{1}{2}\int_t^{t+\tau}(R^m\, dR^n - R^n\, dR^m)$ (Definition 6.1), the stationary Itô identity gives $\mathbb{E}\big[\mathbb{A}^{mn}_{[t,t+\tau]}\big] = -\frac{\tau}{2}(AP - PA^\top)_{mn}$ (exactly, linear in τ), while $\frac{d}{d\tau}\mathcal{A}_\tau\big|_0 = \frac{1}{2}(AP - PA^\top)$; the same stationary current $AP - PA^\top$ thus fixes both (up to sign), and $\mathcal{A}_\tau \equiv 0$ iff the reservoir is reversible.*

*Thus the strong effect vanishes without the area, is linear in it, and — since Q supplies the sign-folding and $\mathcal{A}_\tau$ the arrow of time — requires **both** a quadratic (sign-folding) readout and a genuine price→vol channel.*

Proof. Centre R; the constants $a, \operatorname{tr}(QP)$ drop from covariances, and the linear×quadratic cross-term is a third moment of a mean-zero Gaussian, hence 0 (Lemma 9.0). For the quadratic×quadratic term use the Wick identity for two quadratic forms, $\mathrm{Cov}(X^\top MX,\, X^\top KX) = 2\,\mathrm{tr}(M\Sigma K\Sigma)$ for $X \sim \mathcal{N}(0,\Sigma)$ and symmetric M, K. Stacking $X = (R_{t+\tau}, R_t)$ with $\Sigma = \left(\begin{smallmatrix} P & D_\tau \\ D_\tau^\top & P \end{smallmatrix}\right)$ and selecting the future-variance / past-trend^2 blocks gives $\mathrm{Cov}(R_{t+\tau}^\top QR_{t+\tau}, (u^\top R_t)^2) = 2\,u^\top D_\tau^\top QD_\tau u$; the reversed pairing gives $2\,u^\top D_\tau QD_\tau^\top u$; subtracting yields the first display, and the $S/\mathcal{A}$ form is algebra. For (i)–(ii): reversibility $\iff$ $AP = PA^\top$ $\iff$ D_τ symmetric $\iff$ $\mathcal{A}_\tau \equiv 0$ (Prop. 9.5); the area-rate identity is the stationary Itô computation $\mathbb{E}[R_s^m\, dR_s^n] = (PA^\top)_{mn}\, ds$ (the martingale part has zero mean, so only the drift $\mathbb{E}[R^m(AR)_n] = (PA^\top)_{mn}$ survives). □

Proposition 9.8 (a closed-form two-factor instance). *Take the minimal non-normal reservoir with a Z-driven trend factor feeding a W-driven variance factor at rate γ, at zero leverage ($Z \perp W$):*

$$dR_t^1 = -\lambda R_t^1\, dt + dZ_t, \qquad dR_t^2 = -\mu R_t^2\, dt + \gamma R_t^1\, dt + dW_t, \qquad A = \begin{pmatrix} -\lambda & 0 \\ \gamma & -\mu \end{pmatrix}, \tag{9.26}$$

*non-normal for $\gamma \neq 0$. Here λ is the mean-reversion rate of the price-driven **trend** factor R^1 (so $1/\lambda$ is its trend look-back horizon), μ that of the **variance** factor R^2 (so $1/\mu$ is the volatility-persistence / clustering timescale), and γ is the **trend→variance transmission rate** — the directed, time-irreversible coupling that generates the area (with $\gamma = 0$ the two factors decouple and the effect vanishes). With trend $T = R^1$ ($u = e_1$) and variance readout $V = a + q\,(R^2)^2$ ($Q = q\, e_2 e_2^\top$), the stationary covariance is $p_{11} = \frac{1}{2\lambda}$, $p_{12} = \frac{\gamma}{2\lambda(\lambda+\mu)}$, $p_{22} = \frac{1}{2\mu} + \frac{\gamma^2}{2\lambda\mu(\lambda+\mu)}$, and the symmetric and antisymmetric (area) parts of the lagged $(2,1)$ covariance are*

$$S_{21}(\tau) = \frac{\gamma\,(\lambda e^{-\mu\tau} - \mu e^{-\lambda\tau})}{2\lambda(\lambda^2 - \mu^2)}, \qquad \mathcal{A}_{21}(\tau) = \frac{\gamma}{2}\,\frac{e^{-\mu\tau} - e^{-\lambda\tau}}{\lambda^2 - \mu^2}, \tag{9.27}$$

so that the diagnostic is, in closed form,

$$\mathcal{Z}_{\mathrm{strong}}(\tau) = 2q\big(D_{21}^2 - D_{12}^2\big) = 8q\,\mathcal{A}_{21}(\tau)\, S_{21}(\tau) = \frac{2q\gamma^2}{\lambda(\lambda+\mu)^2}\,\tau + O(\tau^2). \tag{9.28}$$

It is nonzero iff both $\gamma \neq 0$ (the area is present) and $q \neq 0$ (sign-folding is present); its sign is $\mathrm{sign}(q)$*; and it encodes the asymmetry directly: for $\gamma > 0$, $D_{21} = \mathbb{E}[R_{t+\tau}^2 R_t^1] > D_{12} = \mathbb{E}[R_{t+\tau}^1 R_t^2]$, i.e. the past trend forecasts future variance more than the reverse. Here the area is dynamically generated by the non-normal coupling γ, so $\mathcal{Z}_{\mathrm{strong}} \propto \gamma^2$; an explicit area **state** coordinate loaded with weight β — the area-loading weight of Conjecture 9.6 and §6, which the note reserves for this role (§9.1) and which is **not** the Gaussian-readout coefficients a, b, Q used here — would instead, via the bilinear route of §6, make the dependence manifestly linear in β, matching Conjecture 9.6 verbatim.*

Corollary 9.9 (the exponential / Bergomi readout gives the same area-linear effect). *For the exponential readout $V_t = \xi_0 \exp\big(b^\top R_t - \frac{1}{2} b^\top Pb\big)$ (§5, Prop. 5.3 — the Bergomi corner) and the trend $T_t = u^\top R_t$, the Gaussian identity $\mathrm{Cov}(e^G, H^2) = e^{\frac{1}{2}\mathrm{Var}\, G}\, \mathrm{Cov}(G,H)^2$ for jointly Gaussian (G,H) gives the closed form*

$$\mathcal{Z}_{\mathrm{strong}}^{\exp}(\tau) = \xi_0\big[(b^\top D_\tau u)^2 - (b^\top D_\tau^\top u)^2\big] = 4\,\xi_0\,(b^\top \mathcal{A}_\tau u)(b^\top S_\tau u). \tag{9.29}$$

*This is exactly Proposition 9.7 with the effective quadratic $Q = \frac{1}{2} bb^\top$ and prefactor ξ_0: the higher shuffle powers of the exponential resum into the level ξ_0, leaving the same area-linear structure. Hence the reversible-vanishing (C1) and the area-linearity with definite sign (C2) hold **verbatim for the lognormal (Bergomi) readout**, not only the quadratic one — the strong-Zumbach effect* *is a property of the reservoir's non-normality (the area $\mathcal{A}_\tau$), shared by every sign-folding readout, whether polynomial or exponential.* $\square$

What is proved, and what remains. Propositions 9.7–9.8 settle the two structural claims of Conjecture 9.6 — the reversible-vanishing *(C1)* and the area-linearity with explicit sign *(C2)* — in the linear-Gaussian corner — for the quadratic readout, and (Corollary 9.9) for the exponential / Bergomi readout — and identify the kernel $g(\tau)$ with the antisymmetric (area) part $\mathcal{A}_\tau$ of the lagged covariance. They also sharpen Prop. 9.5: the same current $AP - PA^\top$ that measures irreversibility also fixes the expected Lévy-area rate (up to sign; part (ii)) and is thereby the source of the effect. Three gaps remain to the full conjecture. **(a)** The manifestly β-linear statement needs an explicit area *state* coordinate $\mathbb{A}^{Z,n}$ (a level-2, non-Gaussian reservoir), whose moments are still Wick-computable from the underlying Gaussian but no longer through the plain covariance D_τ. **(b)** The quantitative EGRR scaling $\mathcal{Z} \sim \delta^{2H+1}$ requires the rough / multi-scale limit ($N \to \infty$ with the log-grid loadings of Corollary 3.3) together with the short-window δ-expansion — the genuinely open analytic step. **(c)** Transporting the result through a nonlinear (state-dependent-Σ) readout via the perturbative argument of §3.4. Steps (a)–(c) are analysis, not algebra: the finite-dimensional identities above are elementary and have been checked in a computer-algebra system; step (b) is a stochastic-asymptotics problem outside the reach of current formal-proof tooling.

9.3 Does the ESN beat the standard signature for the (strong) Zumbach effect?

> *This subsection synthesises the note's position. The universality facts are CGGOT / Cirone et al. (cited); Prop. 9.5 is the author's; the genericity observation and the comparison itself are the author's framing; the leading-order area-$\Rightarrow$-strong-effect link is Conjecture 9.6 (open).*

The honest answer has three parts, and the first guards against the obvious overclaim.

1. There is no *expressivity* advantage — there cannot be. The untruncated signature is universal: linear functionals of $\mathbb{S}(\widehat{W})$ are dense in continuous path functionals, and every time-irreversible (Lévy-area / lead-lag) term that generates Zumbach asymmetry is *already a coordinate of the signature* — the leading one, the area $\mathbb{A}^{Z,n}$, is a **level-2** coordinate. The randomized signature / ESN is a finite-N random *projection* of that object, universal only as $N \to \infty$ with $O(N^{-1/2})$ error (Theorem 3.6). So in representational power ESN $\subseteq \overline{\text{signature}}$: an ESN cannot produce a strong-Zumbach functional the signature cannot, and at finite N it produces a controlled *approximation* of one. Any claim of the form "ESNs generate the Zumbach effect, signatures do not" is false — both do, the signature exactly.

2. The advantages are *parsimony, state, and genericity*, not power.

- *Dimension — but only for the high-order content.* The leading area term is level-2 and cheap in **both** representations, so for the leading effect there is no dimensional gap. A realistic path-dependent-volatility functional (Guyon-style), however, is a genuinely nonlinear function of several trend features with weight on **high** signature levels; the truncated signature has dimension $(d^{M+1}-1)/(d-1)$, exponential in the level M, whereas the ESN packs a projection of *all* levels into a fixed N-dimensional state. For the full functional — not merely its quadratic leading term — this compression is the real practical win, and is the CGGOT raison d'être.
- *A finite-dimensional Markov state.* Pricing, the VIX conditional expectation (Prop. 9.2), and pathwise Greeks all want a low-dimensional Markov carrier; the ESN is one by construction, the truncated signature only at the exponential dimension above.

- *Genericity buys the irreversibility for free — the sharp point, tied to Prop. 9.5.* The Zumbach asymmetry is carried by the antisymmetric/area part of the dynamics, governed by the **non-normality** of the drift ($[A, A^\top] \neq 0$) and the commutators $[B_i, B_j]$ (defined and disentangled in the block preceding Prop. 9.5). A *generic random* reservoir (CGGOT) is non-normal almost surely, so it **automatically** carries the time-irreversible features. The failure mode is the *non-generic* choice of a **diagonal** (hence normal *and* mutually commuting) reservoir — exactly the Pallavicini architecture — which by Prop. 9.5 is time-**reversible** and yields **no** zero-leverage Zumbach at all. This is theorem-backed from the SSM side: Cirone et al. (2024) prove diagonal selective SSMs are strictly less expressive than non-diagonal ones, with the gap precisely in the lead-lag / parity (i.e. *irreversible*) functions. So within the ESN family the statement is sharp: **you must use a non-normal reservoir — a diagonal one provably cannot produce the effect — and a generic random (or designed non-normal) one does.**

3. What the signature does *better*, and how the structured ESN recovers it. The signature has one genuine advantage the *random* ESN gives up: **clean algebraic control of the asymmetry.** Through the antipode (time reversal) and the shuffle Hopf structure, the irreversible part of the signature is *exactly* the antisymmetric (area) component (Lemma 6.2) — one knows precisely which coordinates are odd under time reversal and can load exactly those. In a generic random reservoir that same information is smeared across all N coordinates in a basis-dependent way. This is exactly the purpose of the **structured / designed-coordinate** route of §6(b): choosing the B_i so that a handful of reservoir coordinates *are* the named area words $\mathbb{A}^{Z,n}$ recovers the signature's interpretability *inside* the low-dimensional, Markov, reusable ESN. The structured ESN is therefore the deliberate interpolation — the parsimony and state of a reservoir with the algebraic transparency of a signature.

Scope caveat (do not skip). Everything above concerns loading the area between the **exogenous** price-driver Z and the volatility channels — a *reusable proxy* for the strong Zumbach effect, in which volatility is a functional of the driving Brownian Z, not of the realised price path. The *genuine* strong effect — volatility as a functional of the realised signed-return path — requires **price-driving** the reservoir (the self-referential / self-consistent construction of §7(c)), which **both** frameworks must adopt equally and which **breaks** the simulate-once reusability. In that genuine-PDV regime the ESN holds no advantage over the signature; the advantage is specific to the reusable-proxy regime. And the claim that loading the area yields the strong-effect diagnostic at leading order remains **Conjecture 9.6 — open**.

One-line summary. Against the *full* signature, the ESN trades a little representational power for a low-dimensional Markov state and — with non-normal/generic dynamics — *free* time-irreversibility; against a *diagonal* ESN, the non-normal/random reservoir provably wins on Zumbach; and against naïveté, the strong effect is only a reusable *proxy* unless one price-drives, with the leading-order link still conjectural.

9.4 Memory structure of the variance

The benchmark also settles *how* the model remembers. The reservoir state is an exponentially-weighted integral of its own past,

$$R_t = \int_{-\infty}^{t} e^{A(t-s)}\, \Sigma\, dW_s, \tag{9.30}$$

so although R is Markov, every observable $V_t = g(R_t)$ is a **non-Markov process whose memory is carried by the kernel** $e^{A(t-s)}$ — the fading-memory / echo-state property. All second-order memory therefore passes through the single kernel autocovariance $C_\tau = Pe^{A^\top \tau} = \sum_n (\mathrm{mode}_n)\, e^{-\kappa_n \tau}$, whose timescales are the reservoir's mean-reversion rates $\kappa_n = -\mathrm{Re}\,\lambda_n(A)$; the readout only decides how those timescales are combined.

Proposition 9.10 (memory structure of the variance). *Let R be the stationary linear-Gaussian reservoir with rates $\kappa_n = -\mathrm{Re}\,\lambda_n(A)$ and $\kappa_{\min} = \min_n \kappa_n$. Each readout's variance autocovariance is a functional of the one kernel autocovariance $C_\tau = Pe^{A^\top \tau}$,*

$$\underbrace{b^\top C_\tau b}_{\text{linear}}, \qquad \underbrace{b^\top C_\tau b + 2\,\mathrm{tr}(QC_\tau QC_\tau^\top)}_{\text{quadratic (Prop. 9.3)}}, \qquad \underbrace{\xi_0^2\big(e^{c(\tau)} - 1\big), \quad c(\tau) = b^\top C_\tau b}_{\text{exponential (Prop. 5.3)}}. \tag{9.31}$$

Then, the stationary mode weights being positive:

(i) ***Long-lag persistence is readout-independent.*** *If the level loading $b \neq 0$, all three decay at the slowest rate as $\tau \to \infty$, $\mathrm{Cov}(V_t, V_{t+\tau}) \asymp e^{-\kappa_{\min}\tau}$ (up to an oscillatory factor for complex modes): the longest memory timescale is $1/\kappa_{\min}$, fixed by the reservoir spectrum, not by the readout.*

(ii) ***The magnitude channel remembers for half as long.*** *The quadratic term $2\,\mathrm{tr}(QC_\tau QC_\tau^\top)$ decays at the summed rates $\{\kappa_m + \kappa_n\}$; for a pure magnitude readout ($b = 0$) the whole autocovariance decays at $2\kappa_{\min}$. The exponential resums all summed-rate channels $\{\sum_j \kappa_{n_j}\}$ into $e^{c(\tau)}$, but its leading term is again $c(\tau)$, so (i) is unaffected. In particular $\mathrm{Cov}(\log V_t, \log V_{t+\tau}) = c(\tau) = b^\top C_\tau b$ exactly.*

(iii) ***Short-lag roughness is a separate feature.*** *For fixed N the structure function $m_2(\tau) = \mathbb{E}[(V_{t+\tau} - V_t)^2] = 2\big(\mathrm{Var}\,V - \mathrm{Cov}(V_t, V_{t+\tau})\big)$ is $\sim C\,\tau$ as $\tau \downarrow 0$ (infinitesimal $H = \frac{1}{2}$), with the pre-asymptotic τ^{2H} window over $[\kappa_{\max}^{-1}, \kappa_{\min}^{-1}]$ supplied by the fractional loading of Cor 3.3.*

The identities are elementary Gaussian-moment computations (checked in a computer-algebra system and by Monte Carlo).

Four consequences worth stating.

- **Persistence and arrow-of-time are governed by different features of A.** The autocovariance above is symmetric in $t \leftrightarrow t + \tau$ and depends only on the **spectrum** $\{\kappa_n\}$ — this is the *persistence* memory. The time-**asymmetric** memory (past trends forecasting future variance more than the reverse, §9.2) is governed instead by the **non-normality** of A — the area $\mathcal{A}_\tau$, i.e. $AP \neq PA^\top$ — and needs a sign-folding readout (Cor 9.9). Spectrum $\Rightarrow$ *how long*; non-normality $\Rightarrow$ *which direction in time.*
- **Roughness and long memory are decoupled — the multi-factor advantage.** A single-parameter fractional driver ties the small-lag roughness and the large-lag persistence to one exponent H; here the *fast* end of the spectrum (with the Cor 3.3 loading) sets the short-lag roughness (iii) while the *slow* end $\kappa_{\min}$ sets the long-memory timescale (i), and the two can be dialled independently. This is precisely the flexibility behind the "rough vs. path-dependent vs. Markovian" debate (Abi Jaber–Li; Rømer): a multi-scale Markov reservoir can carry a rough model's short-lag scaling and a long-memory model's slow ACF at once.
- **Memory is calibrated through the reservoir, not the readout.** Since the timescales are the κ_n, matching an empirical volatility autocorrelation is a statement about the *spectrum* $\{\kappa_n\}$ and the loadings — fixed or randomly drawn in the reservoir-computing philosophy — leaving the trained readout to fit the marginal, the skew, and the level. This is the memory-side companion to the identifiability remark after Prop. 9.3 (level and magnitude channels separate by decay rate).

- **For the exponential (Bergomi) readout the natural memory object is the log-variance.** By (ii) the log-variance carries the raw reservoir memory $c(\tau) = b^\top C_\tau b$ *undistorted*, while the variance memory is its lognormal image $\xi_0^2(e^{c(\tau)} - 1)$. This is exactly why Bergomi and rough-Bergomi specify the dynamics — and the fractional kernel — at the log-variance level: there the memory is linear-Gaussian and the multi-scale / rough kernel of §3.4 enters unmodified.

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10. What this buys, in one paragraph

Reading the signature volatility models as ESNs is exact (Theorem 3.2, Lemma 3.1): the exact-signature/linear-readout model is the state-affine corner, CGGOT is its random projection, and the Volterra kernels that appear in the universality proofs are literally the exponential resummation of the truncated signature (3.4)–(3.5). The ESN viewpoint then supplies three extensions with theory attached: **(a)** a quadratic readout manufactures the sign-folding second-order Volterra kernel and, when the price direction Z is in the control, a *reusable proxy* for the strong Zumbach effect from a *fixed exogenous* reservoir — at the cost of affine pricing (§5, §9.3; the asymmetry itself still needs a price→vol channel, Prop. 9.4), while its **exponential resummation** lands the same construction on the (rough) **Bergomi** model — a lognormal variance sharing the *identical* area-Zumbach mechanism (Corollary 9.9) but trading the affine VIX^2 for a lognormal-basket VIX and a more delicate martingale property (§5, Prop. 5.3); **(b)** structured vector fields let one *place* the time-irreversible Lévy-area coordinates that carry the asymmetry (Lemma 6.2), interpolating between Cuchiero's randomness and a named stylised fact; **(c)** the RCDE umbrella makes "exogenous vs price-driven" a single dial along the fidelity–reusability axis. The natural theoretical target (T3) is an ESN analogue of the EGRR δ^{2H+1} Zumbach formula, with the linear-Gaussian benchmark (T6) and the non-normality/irreversibility unification (T4) as the stepping stones.

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Note. Quantitative claims attached to T3/T4/T5 are conjectural pending the linear-Gaussian computation (T6); the equivalence results (T1) and the universality/rate statements (Theorems 3.5–3.8) are established in the cited literature. The rough-volatility / Volterra-process references supporting §3.4 are standard and were checked for authors, title, year, and venue; exact volume/page numbers for a few (Carmona–Coutin–Montseny, Bayer–Breneis, Rømer, Gassiat) should be verified against the originals.